BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 5 Note Sheets.

Urinary · Male and female reproductive · Brain and brainstem · Meninges and CSF · Nervous tissue · Spinal cord · Peripheral, cranial and autonomic nerves

PART A

The brain dump prompts

All 61 Module 5 competencies, numbers 133 to 193, with two drawing prompts each. Read this part once, then pick.

PART B

The note sheets

One sheet per competency. The competency and its two prompts on the left, one big drawing box on the right. Sixty one sheets.

How this packet works

Every Module 5 competency is on one sheet with two prompts beside one open box. The number on each sheet is the same number it has in your competency packet. You build one prompt in the box. The other is there if you want more practice, and it is a good place to go the week before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before any video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is the point. It is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing. Some want a table. Some want two views side by side. The division is yours to make.
4. Words are allowed inside the drawing, as labels and inside little boxes with arrows between them. A sentence running the width of the box is not. If a box reads like a paragraph, redo it as a picture.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a table or a sequence, so a student who thinks in pictures and a student who thinks in categories both get a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set the same thing up on a tablet, and copy the prompt letter you chose at the top. A hand ruled sheet counts exactly the same as a printed one.

Lecture and lab

Each competency is tagged Lecture, Lab, or both, the same as your competency packet. Lab tagged competencies are on the lab practical, so practice those on the models and the cadaver too.

PART A

The brain dump prompts.

Sixty one competencies, grouped by topic in the order of your competency packet. Each one gives you two prompts. Build one of them in that competency's box on your note sheet in Part B.

The Urinary System

133 Kidney gross anatomy and layers

MODULE 5 · LECTURE · LAB

You should be able to: Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia.

A LABELED LOCATION VIEW

Draw the back wall of the abdomen with the vertebrae from T12 to L3, and place both kidneys with the right one slightly lower. On one kidney, label the renal hilum with the ureter, renal artery, renal vein, and nerves. Around the other, draw and label the renal capsule, adipose capsule, and renal fascia.

Then: Write beside the right kidney why it sits lower, and write beside the left what retroperitoneal tells you about where it sits.

B LAYERS TABLE

Draw a table with the renal capsule, adipose capsule, renal fascia, and renal hilum down the side. Across the top: where it sits (inner, middle, outer, or medial border), what it is made of or what passes through it, and its job. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch both kidneys between T12 and L3 with the right one lower, and mark the hilum on each.

134 Internal kidney regions

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney.

A LABELED FRONTAL SECTION

Draw a kidney cut in a frontal section. Label the renal capsule, renal cortex, renal medulla, the renal pyramids with their bases toward the cortex, the renal columns between them, the renal papillae, minor calyces, major calyces, renal pelvis, and renal sinus. Show the ureter leaving at the hilum.

Then: Shade the parenchyma in one color and write beside it the two regions it includes and what it holds.

B SORTING BOX

Draw two boxes, one labeled parenchyma and one labeled spaces that collect urine. Sort the cortex, renal columns, renal pyramids, renal papillae, minor calyces, major calyces, and renal pelvis into the two boxes. Beside the boxes, draw one pyramid with its papilla draining into a minor calyx, then a major calyx and the renal pelvis.

Then: Label a renal column beside your pyramid and write which region of the kidney it is an extension of.

135 Nephron structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron.

A LABELED NEPHRON

Draw one nephron with its collecting duct. Label the renal corpuscle, the glomerulus, the glomerular capsule with its parietal and visceral layers, the podocytes, the capsular space, the proximal convoluted tubule, the descending and ascending limbs of the nephron loop, the distal convoluted tubule, and the collecting duct.

Then: Draw arrows for the filtrate from the capsular space to the collecting duct, and mark the segment lined with a brush border.

B PARTS TABLE

Draw a table with the glomerulus, glomerular capsule, podocytes, capsular space, proximal convoluted tubule, nephron loop, distal convoluted tubule, and collecting duct down the side, with the renal corpuscle parts first and then the tubule parts in the order the filtrate meets them. Across the top: part of the renal corpuscle or the tubule, and one feature you would use to spot it. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a podocyte wrapped around a glomerular capillary and label its foot processes and the filtration slits.

136 Cortical vs juxtamedullary nephrons

MODULE 5 · LECTURE · LAB

You should be able to: Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds.

A SIDE BY SIDE NEPHRONS

Draw a strip of kidney with the cortex on top, the medulla below, and a line between them. Draw a cortical nephron and a juxtamedullary nephron side by side, showing how deep each nephron loop reaches. Wrap the cortical nephron in peritubular capillaries and the long juxtamedullary loop in the vasa recta.

Then: Write the percent of all nephrons each type makes up beside it, and star the type that concentrates the urine.

B COMPARE TABLE

Draw a table with cortical nephrons and juxtamedullary nephrons down the side. Across the top: share of all nephrons, length of the loop, how far the loop reaches into the medulla, and the capillary bed that serves it. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch one renal pyramid with a short loop and a long loop drawn into it, and label each loop's nephron type.

137 JGA and filtration membrane

MODULE 5 · LECTURE · LAB

You should be able to: Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back.

A LABELED CORPUSCLE CLOSE-UP

Draw a renal corpuscle with its afferent and efferent arterioles and the distal convoluted tubule touching them. Label the juxtaglomerular apparatus, the macula densa cells in the tubule, and the juxtaglomerular (granular) cells in the afferent arteriole. Add a zoomed box of the filtration membrane and label the fenestrated endothelium, basement membrane, and slit membranes.

Then: Draw a blood cell, a large protein, and a medium protein, each stopped at the layer that holds it back.

B WHERE AND WHAT TABLE

Draw a table with the macula densa cells, the juxtaglomerular cells, and the three layers of the filtration membrane down the side, the layers in order from the blood to the capsular space. Across the top: where it sits, and what it senses, releases, or holds back. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the distal convoluted tubule touching the arterioles of its own corpuscle and circle where the juxtaglomerular apparatus forms.

138 Kidney blood supply pathway

MODULE 5 · LECTURE · LAB

You should be able to: Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order.

A LABELED VESSEL CUTAWAY

Draw a frontal section of a kidney showing two pyramids and the cortex. Draw and label the renal artery at the hilum, the segmental arteries branching in the renal sinus, the interlobar arteries running between the pyramids, the arcuate arteries along the pyramid bases, and the interlobular (cortical radiate) arteries up into the cortex. Add one nephron with its afferent arteriole, glomerulus, efferent arteriole, and peritubular capillaries.

Then: Draw the veins in a second shade beside the arteries and label the cortical radiate vein, arcuate vein, and renal vein.

B FLOW CHAIN

Draw a chain of boxes with arrows: renal artery, segmental and interlobar arteries, arcuate artery, interlobular (cortical radiate) artery, afferent arteriole, glomerulus, efferent arteriole, peritubular capillaries and vasa recta, cortical radiate vein, arcuate vein, interlobar vein, renal vein. Under each artery box, write a word or two saying where it runs.

Then: Circle the two capillary beds in your chain and box the one vessel that sits between them.

139 Ureters, bladder, urethra and urine path

MODULE 5 · LECTURE · LAB

You should be able to: Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra.

A LABELED URINARY TRACT

Draw a kidney, ureter, and bladder, with the kidney and bladder cut open. Label the minor and major calyces, renal pelvis, ureter, detrusor muscle in the bladder wall, the trigone with the two ureteral openings and the internal urethral orifice, the internal urethral sphincter, and the external urethral sphincter. Add arrows for urine from the collecting duct to the outside.

Then: Beside the bladder, sketch a male and a female urethra, and label the prostatic, membranous, and spongy parts of the male one.

B FLOW CHAIN

Draw a chain of boxes with arrows for urine: collecting duct, renal papilla, minor calyx, major calyx, renal pelvis, ureter, urinary bladder, urethra. Under the bladder box, list the trigone, detrusor muscle, and both urethral sphincters. Split the urethra box into male (prostatic, membranous, spongy) and female, and write which one is longer.

Then: Write beside each sphincter whether it is smooth or skeletal muscle and whether you control it.

140 Clinical anatomy of urinary disorders

MODULE 5 · LECTURE

You should be able to: Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects.

A DISORDER MAP

Draw the urinary tract from the kidneys to the urethra, with the adipose capsule around one kidney. Write each disorder on the structure it affects: nephroptosis, kidney stones, urinary tract infection, glomerulonephritis, renal failure, and polycystic kidney disease. Draw one stone in the renal pelvis and another lodged in the ureter.

Then: Draw a dropped kidney beside your map with a kinked ureter, and write what loss of fat has to do with it.

B DISORDER TABLE

Draw a table with nephroptosis, kidney stones, urinary tract infection, glomerulonephritis, and polycystic kidney disease down the side. Across the top: the structure it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Below the table, draw a short female urethra beside a long male urethra and write why urinary tract infections are more common in women.

MODULE 5

The Male Reproductive System

141 Scrotum and testis structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules, sustentacular and interstitial cells, rete testis, and efferent ductules.

A LABELED TESTIS SECTION

Draw the scrotum with one testis cut open inside it. Label the scrotal septum, dartos muscle, cremaster muscle, tunica vaginalis, tunica albuginea, lobules, and seminiferous tubules. Add a zoomed circle of one tubule cut across, and label the sustentacular cells in its wall and the interstitial cells between the tubules.

Then: Write beside the dartos and the cremaster which kind of muscle each one is and what it moves.

B STRUCTURE TABLE

Draw a table with the scrotum, dartos muscle, cremaster muscle, tunica vaginalis, tunica albuginea, seminiferous tubules, sustentacular cells, and interstitial cells down the side. Across the top: where it sits (scrotum, covering the testis, or inside the testis) and its job. Use a word or a tiny sketch in each cell.

Then: Circle the row whose cells produce testosterone, and star the two rows that set the temperature of the testes.

142 Sperm duct system and pathway

MODULE 5 · LECTURE · LAB

You should be able to: Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents.

A LABELED DUCT VIEW

Draw a side view of the male pelvis with the testis, the epididymis on it, and the ductus deferens running up out of the scrotum in the spermatic cord. Label the seminal vesicle, the ejaculatory duct, the prostate, and the prostatic, membranous, and spongy urethra. Add arrows for sperm from the seminiferous tubules to the outside.

Then: Draw the spermatic cord cut across as a circle, and label the ductus deferens, blood vessels, nerves, and cremaster muscle inside it.

B FLOW CHAIN

Draw a chain of boxes with arrows: seminiferous tubules, epididymis, ductus deferens, ejaculatory duct, prostatic urethra, membranous urethra, spongy urethra. Under each box, write in a word or two what happens to the sperm there. Box the one structure that produces sperm.

Then: Draw a bracket over the duct that runs inside the spermatic cord, and list the cord's other three contents beside it.

143 Sperm cell structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell.

A LABELED SPERM CELL

Draw one sperm cell large enough to fill the box. Label the head with its nucleus, the acrosome capping the head, the neck, the middle piece, the principal piece, and the end piece. Draw the middle piece packed with small mitochondria and the end piece tapering to a point.

Then: Write beside each part what it holds or does, in three words or fewer.

B PARTS TABLE

Draw a table with the head, acrosome, neck, middle piece, principal piece, and end piece down the side, in order from front to back. Across the top: how you spot it and what it holds or does. Use a word or a tiny sketch in each cell.

Then: Draw a bracket beside the rows that make up the tail, and star the longest part of the tail.

144 Accessory glands and penis

MODULE 5 · LECTURE · LAB

You should be able to: Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis.

A LABELED PENIS VIEW

Draw a side view of the penis and a cross section of its body. On the side view, label the root, body, glans penis, and prepuce. On the cross section, label the two corpora cavernosa on the dorsal side and the corpus spongiosum on the ventral side surrounding the spongy urethra.

Then: Add the seminal vesicles, prostate, and bulbourethral glands as labeled boxes beside your drawing, and write what fluid each adds to semen.

B TWO PART TABLE

Draw a table in two parts. First, put the seminal vesicles, prostate, and bulbourethral glands down the side, with what each adds to semen across the top. Second, put the root, body, glans penis, prepuce, corpora cavernosa, and corpus spongiosum down the side, with what each is. Use a word or a tiny sketch in each cell.

Then: Star the gland that adds the largest share of semen, and circle the erectile column that forms the glans.

145 Clinical anatomy of male disorders

MODULE 5 · LECTURE

You should be able to: Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position.

A DISORDER MAP

Draw a side view of the male pelvis with the testis in the scrotum, the inguinal canal, and the urethra passing through the prostate. Write each disorder on the structure it affects: cryptorchidism, hydrocele, varicocele, benign prostatic hyperplasia, prostate cancer, testicular cancer, and inguinal hernia.

Then: Draw an undescended testis with a dotted line to where it should sit, and write why staying at body temperature harms it.

B DISORDER TABLE

Draw a table with cryptorchidism, hydrocele, varicocele, benign prostatic hyperplasia, and inguinal hernia down the side. Across the top: the structure it affects, where that structure sits, and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch an enlarged prostate squeezing the urethra and write the urinary symptom it causes.

MODULE 5

The Female Reproductive System

146 Ovary structure and follicle stages

MODULE 5 · LECTURE · LAB

You should be able to: Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans.

A LABELED OVARY VIEW

Draw the uterus with one ovary beside it. Label the ovarian ligament, suspensory ligament, broad ligament, and mesovarium. Draw the ovary cut open and label the germinal epithelium, tunica albuginea, cortex, and medulla. In the cortex, draw ovarian follicles, a mature follicle, a corpus luteum, and a corpus albicans.

Then: Number the follicle structures in the order they appear, and mark where the oocyte leaves at ovulation.

B SEQUENCE STRIP

Draw a row of circles showing a follicle growing into a mature follicle, releasing its oocyte, then becoming a corpus luteum and finally a corpus albicans. Beneath the row, list the ovary layers from the outside in, then the three ligaments with what each one attaches the ovary to.

Then: Circle the layer of the ovary where every follicle in your strip is found.

147 Uterine tubes and ovum pathway

MODULE 5 · LECTURE · LAB

You should be able to: Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus.

A LABELED TUBE VIEW

Draw the uterus with one uterine tube and the ovary beside it. Label the fimbriae, infundibulum, ampulla, and isthmus, and show the infundibulum open to the pelvic cavity near the ovary. Draw an oocyte leaving a mature follicle, with arrows following it through the tube into the uterine cavity.

Then: Star where fertilization usually happens, and draw cilia on the tube lining with an arrow showing which way they move the oocyte.

B FLOW CHAIN

Draw a chain of boxes with arrows: ovary, fimbriae and infundibulum, ampulla, isthmus, uterus. Under each box, write what happens to the oocyte there and one feature of that region, such as its shape or width. Use a word or a tiny sketch under each box.

Then: Draw a dotted arrow from the open infundibulum into the pelvic cavity, and write one disorder this open route explains.

148 Uterus regions, ligaments, and wall

MODULE 5 · LECTURE · LAB

You should be able to: Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium.

A LABELED UTERUS VIEW

Draw the uterus cut open from the front, with the uterine tubes and the top of the vagina. Label the fundus, body, cervix, uterine cavity, and cervical canal with the internal and external os. Label the perimetrium, myometrium, and endometrium in the wall. Add the broad, round, uterosacral, and cardinal ligaments as lines running to what each attaches to.

Then: Zoom in on the endometrium and split it into the layer that sheds each month and the layer that rebuilds it.

B REGION STACK AND TABLE

Draw a stack of three boxes from top to bottom for the fundus, body, and cervix, with the uterine cavity and cervical canal drawn inside. Beside it, draw a table with the four ligaments and the three wall layers down the side. Across the top: what it connects to or what it is made of. Use a word or a tiny sketch in each cell.

Then: Star the wall layer that contracts during childbirth, and circle the ligament that runs to the labia majora.

149 Vagina and external genitalia

MODULE 5 · LECTURE · LAB

You should be able to: Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands.

A LABELED VULVA VIEW

Draw the vulva in an inferior view, with the mons pubis at the top. Label the labia majora, labia minora, clitoris, vestibule, urethral orifice, vaginal orifice, hymen, and the greater vestibular glands beside the vaginal opening. Outline the perineum around the genitalia and the anus.

Then: Sketch a side view of the vagina meeting the cervix, and label the fornix around the cervix.

B STRUCTURE TABLE

Draw a table with the vagina, fornix, hymen, mons pubis, labia majora, labia minora, clitoris, vestibule, and greater vestibular glands down the side. Across the top: where it sits and what it is or does. Use a word or a tiny sketch in each cell.

Then: Mark the two structures that are homologous to male structures, and write the male match beside each.

150 Mammary gland structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland.

A LABELED BREAST SECTION

Draw a side view of a breast cut open, sitting over the chest muscles. Label two or three lobes, the lobules inside one lobe, the alveoli inside one lobule, the lactiferous ducts running to the nipple, the areola, and the suspensory ligaments.

Then: Draw arrows for the path of milk from the alveoli to the nipple, and label each structure the arrows pass.

B NESTING DIAGRAM

Draw nested shapes: alveoli inside a lobule, lobules inside a lobe, and the lobe draining by a lactiferous duct to the nipple. Beside it, draw a small table with the nipple, areola, and suspensory ligaments down the side and what each is or does across the top. Use a word or a tiny sketch in each cell.

Then: Write beside your lobe how many lobes one mammary gland has.

The Brain

151 Brain gross organization

MODULE 5 · LECTURE · LAB

You should be able to: Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen.

A LABELED MIDSAGITTAL VIEW

Draw the brain cut down the middle and seen from the side. Label the cerebrum, diencephalon, brainstem, and cerebellum, and outline each one so the four divisions are easy to tell apart. Along the edge of the cerebrum, draw a thin outer band of gray matter over a thick core of white matter, and label both.

Then: Write beside the gray matter band what it is made of, and beside the white matter what it is made of.

B DIVISION TABLE

Draw a table with the cerebrum, diencephalon, brainstem, and cerebellum down the side. Across the top: where it sits and its main job. Use a word or a tiny sketch in each cell. Below the table, draw a small block of brain tissue with gray matter on the outside and white matter on the inside, and label both.

Then: Star the largest division and the second largest division, and write the cranial fossa where the cerebellum sits.

152 Neural tube brain development

MODULE 5 · LECTURE

You should be able to: Trace brain development from the neural tube through the primary and secondary vesicles and match each secondary vesicle to its adult structures and ventricle.

A BRANCHING FLOW CHAIN

Draw the neural tube at the top of the box. Branch it into the three primary vesicles, writing each one's formal name and common name (forebrain, midbrain, hindbrain). Branch those into the five secondary vesicles. Under each secondary vesicle, write the adult structures it becomes and its ventricle or fluid space.

Then: Write the germ layer the neural tube forms from, and circle the one primary vesicle that carries through without splitting.

B VESICLE TABLE

Write neural tube at the top of the box. Below it, draw a table with the five secondary vesicles down the side. Across the top: the primary vesicle it came from, the adult structures it becomes, and its ventricle or fluid space. Use a word or a tiny sketch in each cell.

Then: Draw a simple side view of the adult brain and number each region 1 to 5 to match the secondary vesicle it grew from.

153 Cerebral surface and lobes

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen.

A SIDE AND TOP VIEWS

Draw the left side of the brain. Label the frontal, parietal, temporal, and occipital lobes, a gyrus, a sulcus, the precentral and postcentral gyri, and the central, lateral, and parieto-occipital sulci. Draw the insula as a dashed shape inside the lateral sulcus. Show the transverse fissure above the cerebellum. Beside it, draw a small top view and label both hemispheres and the longitudinal fissure.

Then: Write beside each of the four outer lobes the cranial bone that lies over it and gives it its name.

B BOUNDARY TABLE

Sketch three folds of cortex and label a gyrus, a sulcus, and a fissure. Below, draw a table with the central sulcus, lateral sulcus, parieto-occipital sulcus, longitudinal fissure, and transverse fissure down the side. Across the top: the region on one side of it and the region on the other. Use a word or a tiny sketch in each cell.

Then: Draw the brain seen from above, label the longitudinal fissure, central sulcus, and each lobe you can see, then name the lobe no outside view shows.

154 Functional cortical areas

MODULE 5 · LECTURE · LAB

You should be able to: Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex.

A LABELED CORTEX MAP

Draw the side view of the hemisphere where Broca's and Wernicke's areas are usually found. Draw the central and lateral sulci. Label the primary motor cortex, primary somatosensory cortex, Broca's area, Wernicke's area, prefrontal cortex, primary auditory cortex, and primary visual cortex. Draw the insula dashed for the gustatory cortex, and point an arrow to the medial temporal lobe for the olfactory cortex.

Then: Write M, S, or A on each labeled area to sort it as a motor, sensory, or association area.

B AREA LOCATION TABLE

Draw a table with these areas down the side: primary motor, primary somatosensory, primary visual, primary auditory, primary gustatory, and primary olfactory cortex, Broca's area, Wernicke's area, and the prefrontal cortex. Across the top: lobe, gyrus or landmark, motor, sensory, or association, and job. Use a word or a tiny sketch in each cell.

Then: Add a row for each association area the notes place next to a primary area, and name the primary area it sits beside.

155 Cerebral white matter tracts

MODULE 5 · LECTURE · LAB

You should be able to: Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen.

A LABELED FRONTAL SECTION

Draw a frontal section through both hemispheres with gray cortex on the outside and white matter inside. Draw and label an association fiber arching between two gyri on the same side, the corpus callosum crossing between the hemispheres, and the projection fibers of the internal capsule running down beside the basal ganglia toward the brainstem.

Then: Shade the internal capsule and write beside it why a small stroke there can weaken a whole side of the body.

B FIBER CLASS TABLE

Draw a table with association, commissural, and projection fibers down the side. Across the top: what it connects, a named example, and a tiny arrow sketch of its path. Use a word or a tiny sketch in each cell. Below, sketch a midline view with the corpus callosum and a frontal section with the internal capsule, and label both.

Then: Circle the largest commissural tract, and write which tract links Broca's and Wernicke's areas and which fiber class it belongs to.

156 Basal ganglia

MODULE 5 · LECTURE · LAB

You should be able to: Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus.

A LABELED FRONTAL SECTION

Draw a frontal section of one hemisphere with cortex on the outside, white matter inside, and deep masses of gray matter buried in the white matter. Label the caudate nucleus, the putamen, and the globus pallidus medial to the putamen. Below the hemisphere, sketch the diencephalon and midbrain, and mark the subthalamic nucleus and the substantia nigra.

Then: Draw dotted lines linking all five nuclei, and write the three roles the basal ganglia share.

B LOCATION SORT

Draw three boxes stacked from top to bottom labeled cerebrum, diencephalon, and midbrain. Write each nucleus in the box where it sits: caudate nucleus, putamen, globus pallidus, subthalamic nucleus, and substantia nigra. Circle the three core nuclei and underline the two grouped with them only by function. Add an arrow showing the putamen lateral to the globus pallidus.

Then: Write beside the substantia nigra what it looks like and the disease its loss causes, and give one semi-voluntary movement the basal ganglia control.

157 Diencephalon

MODULE 5 · LECTURE · LAB

You should be able to: Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose.

A LABELED MIDSAGITTAL VIEW

Draw the brain cut down the midline. Label the thalamus with the massa intermedia, the hypothalamus below it, the epithalamus at the back with the pineal gland, and the third ventricle. Add the corpus callosum above, the infundibulum and optic chiasm below the hypothalamus, and the midbrain below as landmarks.

Then: Shade the third ventricle, and write the percent of the diencephalon the thalamus makes up.

B THREE PART TABLE

Draw a table with the thalamus, hypothalamus, and epithalamus down the side. Across the top: where it sits, what it contains, and its main job. Use a word or a tiny sketch in each cell. Below the table, draw the third ventricle as a midline cavity and label it.

Then: Draw the right and left thalami on each side of your third ventricle, and join them across it with the massa intermedia.

158 Cerebellum

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections.

A BACK AND CUT VIEWS

Draw the cerebellum from behind and label the two hemispheres, the vermis, and the folia. Beside it, draw the cerebellum cut down the midline and attached to the brainstem. Label the cerebellar cortex, arbor vitae, a deep cerebellar nucleus, and the superior, middle, and inferior cerebellar peduncles as stalks running to the midbrain, pons, and medulla.

Then: Shade the gray matter on your cut view and leave the white matter blank.

B PARTS TABLE

Draw a table with these down the side: cerebellar hemispheres, vermis, folia, arbor vitae, deep cerebellar nuclei, and the superior, middle, and inferior cerebellar peduncles. Across the top: what it is, gray or white matter, and where it sits or what it joins. Use a word or a tiny sketch in each cell.

Then: Draw an arrow on each peduncle row showing whether it mostly carries signals into the cerebellum or out of it.

159 Limbic system

MODULE 5 · LECTURE · LAB

You should be able to: Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon.

A LABELED MIDSAGITTAL VIEW

Draw the brain cut down the midline. Label the corpus callosum, the cingulate gyrus arching above it, and the diencephalon beneath it. Draw the hippocampus and amygdala as dashed shapes curving down and forward below the diencephalon, so that the three limbic structures form a ring around it.

Then: Write beside each limbic structure its main role from the notes.

B LIMBIC COMPARE TABLE

Draw a table with the cingulate gyrus, amygdala, and hippocampus down the side. Across the top: where it sits, its role, and a tiny sketch. Use a word or a tiny sketch in each cell. Below the table, draw a small oval for the diencephalon and arrange the three structures around it as a ring.

Then: Write in the center of your ring the three jobs of the limbic system as a whole.

160 Cerebral arterial supply

MODULE 5 · LECTURE · LAB

You should be able to: Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches.

A LABELED INFERIOR VIEW

Draw the base of the brain seen from below with the brainstem in the middle. Draw the vertebral arteries joining into the basilar artery, then the ring: posterior cerebral, posterior communicating, internal carotid, anterior cerebral, and anterior communicating arteries. Add the middle cerebral arteries and the cerebellar branches: posterior inferior, anterior inferior, and superior cerebellar arteries.

Then: Shade the anterior circulation one way and the posterior circulation another, and write what each one feeds.

B TWO CIRCULATION CHAINS

Draw two flow chains side by side. Anterior: internal carotid artery to the anterior and middle cerebral arteries. Posterior: vertebral arteries to basilar artery to posterior cerebral arteries, with the posterior inferior cerebellar branch off each vertebral and the anterior inferior and superior cerebellar branches off the basilar. Join the two chains with the posterior communicating artery, and add the anterior communicating artery joining the right and left anterior cerebral arteries.

Then: Write beside each cerebral artery the part of the brain it supplies.

MODULE 5

The Brainstem

161 Brainstem regions

MODULE 5 · LECTURE · LAB

You should be able to: Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord.

A LABELED SIDE VIEW

Draw the brainstem from the side with the diencephalon above it, the cerebellum behind it, and the spinal cord below. Label the midbrain, pons, and medulla oblongata from top to bottom, the basilar pons, and the pontomedullary junction. Mark the foramen magnum where the medulla becomes the spinal cord.

Then: Draw one arrow up and one arrow down through all three regions, and label them ascending sensory tracts and descending motor tracts.

B REGION ORDER CHAIN

Draw a vertical chain of boxes from top to bottom: cerebrum and diencephalon, midbrain, pons, medulla oblongata, and spinal cord, with the cerebellum to the side. In each brainstem box, write its position (most superior, middle, most inferior) and one structure you can find on it on the model.

Then: Write beside the chain what passes through all three regions and how many cranial nerves have nuclei in the brainstem.

162 Midbrain structures

MODULE 5 · LECTURE · LAB

You should be able to: Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen.

A TWO LABELED VIEWS

Draw the midbrain from behind and in cross section. On the back view, label the tectum and its four colliculi, with the superior pair above the inferior pair. On the cross section, label the tectum, the cerebral aqueduct, the darkly pigmented substantia nigra, and the cerebral peduncles on the front.

Then: Draw an arrow into and out of the cerebral aqueduct, and name the ventricle at each end.

B MIDBRAIN STRUCTURE TABLE

Draw a table with the tectum, superior colliculi, inferior colliculi, cerebral peduncles, substantia nigra, and cerebral aqueduct down the side. Across the top: how to spot it on the model, and its job. Use a word or a tiny sketch in each cell.

Then: Circle the two reflex centers, and write beside them whether you become conscious of the sight or sound there.

163 Pons structures

MODULE 5 · LECTURE · LAB

You should be able to: Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it.

A LABELED SIDE VIEW

Draw the brainstem from the left side with the cerebellum behind it. Label the midbrain, pons, and medulla, the basilar pons, and the thick middle cerebellar peduncle running back into the cerebellum. Dot the pontine nuclei inside the pons. Draw the fourth ventricle as a dashed space behind the pons and upper medulla.

Then: Draw an arrow from the cerebral cortex through the pontine nuclei and the middle cerebellar peduncle into the cerebellum.

B PONS FLOW CHAIN

Draw a flow chain for motor information: cerebral cortex, pontine nuclei, middle cerebellar peduncles, cerebellum. Write what each step does in a few words. Under the chain, sketch the pons with the midbrain above, the medulla below, and the fourth ventricle behind it, and label all four.

Then: Write what the word pons means, and name the three regions it bridges.

164 Medulla oblongata structures

MODULE 5 · LECTURE · LAB

You should be able to: Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers.

A LABELED ANTERIOR VIEW

Draw the brainstem from the front with the spinal cord below. Label the pons, the pontomedullary junction, the medulla oblongata, the two pyramids, the medullary olive on each side of the pyramids, and the decussation of the pyramids where the medulla meets the cord. Inside the medulla, draw and label boxes for the cardiac, respiratory, and vasomotor centers.

Then: Draw crossing arrows at the decussation, and write why a stroke in the right hemisphere weakens the left side of the body.

B MEDULLA TABLE

Draw a table with the pyramids, medullary olive, decussation of the pyramids, cardiac center, respiratory center, and vasomotor center down the side. Across the top: seen on the surface or inside the medulla, and what it looks like or what it does. Use a word or a tiny sketch in each cell.

Then: Star the three vital centers, and write why injury to the medulla is immediately life threatening.

165 Cranial nerve nuclei distribution

MODULE 5 · LECTURE · LAB

You should be able to: Match cranial nerve nuclei III through XII to the brainstem region that houses them.

A LABELED BRAINSTEM VIEW

Draw the brainstem from the front, divided into the midbrain, pons, and medulla oblongata, and label the pontomedullary junction. Beside each region, write the Roman numerals and names of the cranial nerves whose nuclei it holds, from CN III down to CN XII.

Then: Write which two cranial nerves have no nuclei in the brainstem, and circle the one nerve you wrote beside two regions.

B NERVE SORT TABLE

Draw a table with CN III through CN XII down the side. Across the top: the nerve's name, midbrain, pons, and medulla oblongata. Write each name, then put a check in the region that houses that nerve's nuclei. Use a word or a check in each cell.

Then: Count how many of the twelve cranial nerves have nuclei in the brainstem, and circle the nerve with checks in two regions.

166 Reticular formation

MODULE 5 · LECTURE

You should be able to: Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component.

A LABELED CUTAWAY

Draw the brainstem from the side, cut open to show its core. Draw the reticular formation as a net of dots and lines woven through the midbrain, pons, and medulla. Draw arrows up from the net to the cerebral cortex labeled reticular activating system, and arrows down toward the spinal cord labeled descending component.

Then: Write beside each set of arrows two of its jobs from the notes.

B COMPARE TABLE

Draw a table with the reticular activating system and the descending component down the side. Across the top: which way it sends signals, what it acts on, and its jobs. Use a word or a tiny sketch in each cell. Above the table, sketch the brainstem core with a gray matter net running through its white matter.

Then: Write under the table what happens when the reticular activating system shuts down and what severe damage to it can cause.

MODULE 5

Meninges and Cerebrospinal Fluid

167 Cranial meninges and dural reflections

MODULE 5 · LECTURE · LAB

You should be able to: Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein.

A LABELED MIDSAGITTAL VIEW

Draw the inside of the skull in midsagittal view with the brain removed. Label the falx cerebri, falx cerebelli, and tentorium cerebelli. Draw the superior sagittal sinus on the falx cerebri, then the confluence of sinuses, a transverse sinus, a sigmoid sinus, and the internal jugular vein. In a corner, draw the dura, arachnoid, and pia from skull to brain.

Then: Add arrows along the sinuses for the direction blood drains, and label the periosteal and meningeal layers on your dura strip.

B TABLE AND FLOW CHAIN

Draw a table with the dura mater, arachnoid mater, pia mater, falx cerebri, falx cerebelli, and tentorium cerebelli down the side. Across the top: where it sits and what it touches or separates. Use a word or a tiny sketch in each cell. Below the table, draw a chain of boxes from the superior sagittal sinus to the internal jugular vein.

Then: Mark the box in your chain where the arachnoid granulations return CSF to the blood.

168 Spinal meninges and meningeal spaces

MODULE 5 · LECTURE · LAB

You should be able to: Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents.

A LABELED CROSS SECTION

Draw a cross section of a vertebra with the spinal cord inside. Label the vertebral bone, the epidural space with fat and veins, the dura mater, subdural space, arachnoid mater, subarachnoid space with CSF, and pia mater. Draw a denticulate ligament from the pia to the dura. Beside it, sketch the lower cord with the filum terminale running to the coccyx.

Then: Star the space that holds CSF, and write potential space beside the space that is only a potential space.

B SPACE COMPARE TABLE

Draw a table with the epidural, subdural, and subarachnoid spaces down the side. Across the top: the two layers it sits between, what fills it, and whether it is also found in the cranium. Use a word or a tiny sketch in each cell. Below the table, sketch the cord with two denticulate ligaments and the filum terminale.

Then: Write beside each anchor which meningeal layer it is made of and what it attaches the cord to.

169 Cord termination and lumbar puncture

MODULE 5 · LECTURE · LAB

You should be able to: Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space.

A LABELED SIDE VIEW

Draw a side view of the lower vertebral column with the spinal cord inside. Label the conus medullaris at the L1 to L2 disc, the cauda equina, the filum terminale, and the lumbar cistern below the cord. Draw a needle entering in the midline between the L3 and L4 spines, and label each layer it crosses, in order, from the skin to the subarachnoid space.

Then: Write beside the conus medullaris why the needle is placed between the L3 and L4 or L4 and L5 spines.

B NEEDLE PATH SEQUENCE

Draw a vertical stack of boxes, one for each layer a lumbar puncture needle crosses, in order from the skin at the top to the subarachnoid space at the bottom. Name the layer and sketch what it looks like in each box. Beside the bottom box, sketch the lumbar cistern with the floating cauda equina, the conus medullaris above it, and the filum terminale.

Then: Mark the box where a clinician feels a firm ligament give way, and write the two spine levels where the needle can enter.

170 Ventricular system

MODULE 5 · LECTURE · LAB

You should be able to: Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them.

A LABELED VENTRICLE VIEW

Draw a side view of the brain outline with the ventricles inside it, as on the brain model. Label a lateral ventricle, an interventricular foramen of Monro, the third ventricle, the cerebral aqueduct, the fourth ventricle, and the central canal of the spinal cord. Draw the median aperture and the two lateral apertures opening from the fourth ventricle into the subarachnoid space.

Then: Write beside each ventricle the region of the brain it sits in, and write what lines every chamber.

B CONNECTION TABLE

Draw a table with the lateral ventricles, third ventricle, fourth ventricle, and central canal down the side. Across the top: where it is, what it connects to next, and the opening or channel that makes the connection. Use a word or a tiny sketch in each cell.

Then: Circle the three narrow connection points in your table, and write beside the table what would happen if one were blocked.

171 CSF production and circulation

MODULE 5 · LECTURE · LAB

You should be able to: Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations.

A LABELED FLOW DRAWING

Draw a midsagittal view of the brain and upper spinal cord showing the ventricles, the subarachnoid space, and the superior sagittal sinus. Draw choroid plexus in the lateral, third, and fourth ventricles. Add numbered arrows for CSF from the lateral ventricles through the interventricular foramina, third ventricle, cerebral aqueduct, fourth ventricle, apertures, and subarachnoid space to the arachnoid granulations.

Then: Add a side arrow into the central canal, and write which cells of the choroid plexus produce the CSF.

B FLOW CHAIN

Draw a chain of boxes with arrows for the one-way loop of CSF, from production at the choroid plexus of the lateral ventricles to reabsorption at the arachnoid granulations. Give each ventricle, opening, and space its own box. Add a tiny sketch in each box of what the structure looks like.

Then: Mark each box that holds choroid plexus, and add a final arrow into the venous blood of the superior sagittal sinus.

172 Blood-brain barrier structure

MODULE 5 · LECTURE · LAB

You should be able to: Name the three structural components of the blood-brain barrier the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet.

A LABELED CAPILLARY SECTION

Draw a cross section of a brain capillary with blood in its center. Label the capillary endothelial cells joined by tight junctions, the basement membrane wrapped around them, and the astrocyte end-feet covering the outside. Draw one whole star-shaped astrocyte with its end-feet reaching the capillary and a neuron nearby.

Then: Draw an arrow from the blood toward the neuron, and number each barrier component the arrow has to cross.

B COMPARE TABLE

Draw a table with the three components of the blood-brain barrier down the side. Across the top: what it is, its position counting outward from the blood, and one feature that makes it part of the barrier. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch an ordinary capillary beside a brain capillary and label the pores on the ordinary capillary that the brain capillary lacks.

MODULE 5

Functional Organization and Nervous Tissue

173 Nervous system divisions

MODULE 5 · LECTURE

You should be able to: Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions.

A BRANCHING MAP

Draw a branching map that starts with the nervous system and splits into the CNS and PNS. Under the CNS, write the brain and spinal cord. Under the PNS, list nerves, ganglia, and sensory receptors, then branch into the somatic, autonomic, and enteric motor divisions. Split the autonomic division into sympathetic and parasympathetic.

Then: Write beside each motor division the effector it controls, and add the number of pairs of cranial and spinal nerves.

B LABELED BODY OUTLINE

Draw a body outline with the brain and spinal cord shaded as the CNS. Draw a few nerves leaving the cord, one ganglion, and a sensory receptor in the skin, and label them as PNS. Draw a skeletal muscle, the heart, a gland, and a loop of gut wall with its own neurons.

Then: Label each effector with the motor division that controls it: somatic, autonomic, or enteric.

174 Neuron structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them.

A LABELED NEURON

Draw a multipolar neuron as it looks on the neuron model. Label the cell body with its nucleus and nucleolus, the Nissl bodies, the dendrites, the axon hillock, the axon, and the axon terminals with their synaptic bulbs.

Then: Add arrows along the neuron for the direction a signal travels, from the dendrites through to the axon terminals.

B SIGNAL FLOW CHAIN

Draw a chain of boxes in the order a signal travels through a neuron: dendrites, cell body, axon hillock, axon, and axon terminals. In each box, sketch the part and write its job in a word or two. Draw Nissl bodies inside the cell body box and write what they are made of and what they do.

Then: Add neurofibrils to the cell body box, and write why an axon dies when it is cut away from its cell body.

175 Synapse structures

MODULE 5 · LECTURE · LAB

You should be able to: Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact.

A LABELED SYNAPSE CLOSE-UP

Draw the end of an axon meeting a second neuron. Label the axon terminal, a synaptic end bulb, the synaptic vesicles inside it, the synaptic cleft, and the next cell. Draw neurotransmitter molecules leaving the vesicles and crossing the cleft. Beside it, draw three tiny sketches of an axon ending on a dendrite, on a cell body, and on another axon.

Then: Name each of your three tiny sketches by the site it contacts: axodendritic, axosomatic, or axoaxonic.

B SYNAPSE TYPE MAP

Draw one neuron with three axons from other neurons ending on it: one on a dendrite, one on the cell body, and one on its axon. Name each synapse by the site it contacts. Draw a zoom circle on one synapse showing the synaptic end bulb, synaptic vesicles, and synaptic cleft.

Then: Write beside the zoom circle what the vesicles store and what carries the signal across the cleft.

176 Neuron classification

MODULE 5 · LECTURE · LAB

You should be able to: Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each.

A THREE SHAPE SKETCHES

Draw a multipolar, a bipolar, and a unipolar neuron side by side. Label the dendrites, cell body, and axon on each one, and write how many processes leave the cell body. Under the multipolar and bipolar shapes, write where each is found, and add the Purkinje cell and pyramidal cell under the shape they belong to.

Then: Label which shape most sensory neurons have, and which shape motor neurons and interneurons have.

B FUNCTIONAL PATHWAY MAP

Draw a skin receptor, a cross section of the spinal cord, and a skeletal muscle. Connect them with a sensory neuron carrying a signal toward the CNS, an interneuron inside the cord, and a motor neuron carrying a command out to the muscle. Draw each neuron in its structural shape and label the shape.

Then: In the margin, add a bipolar neuron and name the three places it is found.

177 Neuroglia identification

MODULE 5 · LECTURE · LAB

You should be able to: Name the four CNS glia and two PNS glia and describe the support role each one performs.

A LABELED GLIA SCENE

Draw a patch of CNS tissue with a neuron, a capillary, and the edge of a ventricle. Draw and label an astrocyte touching the capillary, an oligodendrocyte wrapping several axons, a microglial cell eating debris, and ependymal cells lining the ventricle. In a separate PNS box, draw Schwann cells on an axon and satellite cells around cell bodies in a ganglion.

Then: Write the support role of each glial cell beside its label in a few words.

B SORTING BOX

Draw a box split into a CNS side and a PNS side. Sort the six glial cells into the correct side. For each cell, add a tiny sketch of its shape and a few words for the support role it performs.

Then: Circle the two cells that make myelin, and star the cell that helps form the blood-brain barrier.

178 Myelination and regeneration

MODULE 5 · LECTURE · LAB

You should be able to: Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential.

A LABELED AXON COMPARISON

Draw a PNS axon wrapped by a row of Schwann cells. Label the myelin sheath, the nodes of Ranvier, and the neurolemma. Beside it, draw one CNS oligodendrocyte with arms wrapping segments of several different axons, and label its myelin sheath and nodes of Ranvier. Show that the CNS axons have no neurolemma.

Then: Draw a cut across each axon, then write which one can regrow and the reason beside it.

B COMPARE TABLE

Draw a table with PNS and CNS across the top. Down the side: the cell that makes the myelin, how many axons one cell wraps, whether a neurolemma is present, and whether a cut axon can regrow. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a myelinated axon, write what myelin is made of, and draw the impulse jumping from node to node.

MODULE 5

Gross Anatomy and Neuronal Integration

179 CNS and PNS tissue terms

MODULE 5 · LECTURE

You should be able to: Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS.

A TWO BY TWO GRID

Draw a grid with CNS and PNS across the top. Down the side: a cluster of cell bodies and a bundle of axons. Write nucleus, ganglion, tract, and nerve in the correct boxes. Add a tiny sketch in each box, with dots for cell bodies and lines for axons.

Then: Write a one-line definition under each term that uses the words cluster or bundle and CNS or PNS.

B LABELED NERVOUS SYSTEM MAP

Draw the brain and spinal cord with one nerve leaving the cord and a ganglion along that nerve outside the CNS. Draw a nucleus as a cluster of dots inside the brain and a tract as a bundle of lines running up the cord. Label all four structures.

Then: Draw a dashed line where the CNS ends and the PNS begins, and label each structure cell bodies or axons.

180 Gray and white matter

MODULE 5 · LECTURE · LAB

You should be able to: Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord.

A TWO LABELED SECTIONS

Draw a spinal cord cross section and a slice through the cerebrum side by side, as on the models. Shade gray matter one way and white matter another. On the brain, label the cortex and the white matter beneath it. On the cord, label the gray matter with its dorsal and ventral horns, the outer white matter, and the central canal.

Then: Add a small cerebellum with its gray cortex, and write what gray and white matter are made of in your shading key.

B COMPARE TABLE

Draw a table with gray matter and white matter down the side. Across the top: what it is made of, where it sits in the brain, and where it sits in the spinal cord. Use a word or a tiny sketch in each cell.

Then: Below the table, draw two circles labeled brain and cord, and shade each to show which matter is on the outside.

181 Reflex arc components

MODULE 5 · LECTURE

You should be able to: Name the five components of a reflex arc in order from receptor to effector and trace a signal through them.

A LABELED REFLEX ARC

Draw a receptor, a cross section of the spinal cord, and a muscle. Connect them with a sensory neuron running into the cord, a synapse in the cord, and a motor neuron running out to the muscle. Label the five components of the reflex arc and add arrows for the direction of the signal.

Then: Put an X at one spot on the arc and write what a missing reflex tells a clinician about that pathway.

B FLOW CHAIN

Draw a chain of five boxes for the reflex arc in order, from receptor to effector. In each box, name the component, sketch it, and write its job in a few words. Draw an arrow between each pair of boxes for the direction of the signal.

Then: Circle the box where the synapses are, and write beside the chain why a reflex is fast.

182 Types of reflexes

MODULE 5 · LECTURE

You should be able to: Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each.

A COMPARE TABLE

Draw a table with six reflex types down the side: somatic, autonomic, monosynaptic, polysynaptic, innate, and learned. Across the top: what the name is based on and one example. Use a word or a tiny sketch in each cell.

Then: Draw a bracket joining each pair of types, and write the one feature that splits that pair.

B PAIRED SKETCHES

Draw three pairs of small sketches. For somatic and autonomic, show the effector each one uses. For monosynaptic and polysynaptic, show one synapse in the cord beside a chain with interneurons. For innate and learned, sketch one example of each. Label every sketch with its reflex type and its example.

Then: Write under each pair what the names are based on: the effector, the number of synapses, or whether it is inborn.

183 Neuronal pools and pathways

MODULE 5 · LECTURE

You should be able to: Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation.

A LABELED CIRCUIT DRAWING

Draw a convergence circuit with many neurons feeding into one, and a divergence circuit with one neuron spreading to many, with arrows. Below them, draw a front view of the brain and spinal cord with one ascending and one descending pathway. Show each pathway crossing the midline and label the crossing decussation.

Then: Write beside the brain why a stroke on one side of the brain causes weakness on the opposite side.

B COMPARE TABLE

Draw a table with convergence, divergence, ascending pathway, descending pathway, and decussation down the side. Across the top: what it is and which way the signal travels. Use a word or a tiny sketch in each cell.

Then: Below the table, draw a small body outline and shade the half the left side of the brain controls.

The Spinal Cord

184 Spinal cord external anatomy

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus.

A LABELED POSTERIOR VIEW

Draw the whole spinal cord from the back, running from the medulla down to its tapered end, with the vertebral column drawn faintly behind it. Label the cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, filum terminale, and posterior median sulcus. Mark the disc where the cord ends.

Then: Draw a small cross section beside it, label the anterior median fissure and posterior median sulcus, and show which groove is deeper.

B STRUCTURE AND LANDMARK TABLE

Draw a table with seven structures down the side: cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus. Across the top: what it looks like, where it sits on the cord, and what it does or serves. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the lower cord and mark where a lumbar puncture needle goes, then write why that spot does not risk the cord.

185 Roots, rootlets, and root ganglion

MODULE 5 · LECTURE · LAB

You should be able to: Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons.

A LABELED CUTAWAY

Draw a cross section of the spinal cord with a row of rootlets leaving the back and the front on one side. Gather them into the posterior root and the anterior root, add the posterior root ganglion, and join the two roots into a spinal nerve. Label each part and the intervertebral foramen where the nerve exits.

Then: Write sensory or motor beside each root, and add arrows showing signals coming in on one root and going out on the other.

B FLOW CHAIN

Draw two chains of boxes with arrows. The sensory chain runs from a receptor in the skin through the spinal nerve, posterior root, and rootlets into the cord. The motor chain runs from the cord through the rootlets, anterior root, and spinal nerve out to a muscle. Add the posterior root ganglion and intervertebral foramen where they belong.

Then: Circle the box that holds the sensory cell bodies, and star the one nerve both chains travel in and write why it is called mixed.

186 Spinal cord internal anatomy

MODULE 5 · LAB

You should be able to: Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns.

A LABELED CROSS SECTION

Draw a cross section of a thoracic segment of the cord as it looks on the slide or model, with the gray butterfly inside the white matter. Label the posterior, lateral, and anterior horns, the gray commissure, the central canal, the anterior white commissure, and the anterior, lateral, and posterior white columns.

Then: Write beside each horn, in a few words, what it holds, and write tracts inside one white column.

B LANDMARK TABLE

Draw a table with these down the side: the posterior, lateral, and anterior horns, the gray commissure, the anterior white commissure, the central canal, and the anterior, lateral, and posterior columns. Across the top: gray or white matter, how to find it on the section, and what it holds or connects. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a cervical and a thoracic section side by side and circle the lateral horn on the one that has it.

The Peripheral Nervous System

187 Nerve connective tissue wrappings

MODULE 5 · LECTURE · LAB

You should be able to: Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps.

A LABELED NERVE CROSS SECTION

Draw a cross section of a peripheral nerve as it looks on the slide. Show the whole nerve with several fascicles inside it, then enlarge one fascicle so you can see a few axons. Label an axon, a fascicle, the endoneurium, the perineurium, and the epineurium.

Then: Color each wrapping a different shade and write beside each one the level it wraps: one axon, one fascicle, or the whole nerve.

B WRAPPING TABLE

Draw a table with the three wrappings down the side: endoneurium, perineurium, and epineurium. Across the top: inner, middle, or outer layer, the level it wraps (axon, fascicle, or whole nerve), and a tiny sketch of that level with the wrapping drawn around it. Use a word or a tiny sketch in each cell.

Then: Below the table, write how the perineurium and epineurium help a surgeon repair a cut nerve.

188 Spinal nerves and rami

MODULE 5 · LECTURE · LAB

You should be able to: State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies.

A LABELED SPINAL NERVE

Draw a cross section of the spinal cord with one spinal nerve leaving through the intervertebral foramen. Show it splitting into the posterior ramus, the anterior ramus, and the meningeal branch, with the rami communicantes linking the anterior ramus to a sympathetic trunk ganglion. Label each branch and draw an arrow toward the territory it supplies.

Then: Beside the cord, write the number of spinal nerve pairs in each region, cervical through coccygeal, and the total.

B COUNT AND TERRITORY TABLE

Draw a table in two parts. In the top part, list the five cord regions down the side with the number of spinal nerve pairs in each, and the total. In the bottom part, list the posterior ramus, anterior ramus, meningeal branch, and rami communicantes, with the territory each supplies. Use a word or a tiny sketch in each cell.

Then: Star the ramus that forms the plexuses, and write the name and spinal levels of the thoracic nerves that do not form one.

189 Sensory receptor classification

MODULE 5 · LECTURE · LAB

You should be able to: Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus.

A LABELED SKIN BLOCK

Draw a block of skin (epidermis, dermis, hypodermis) with one hair, and a skeletal muscle with its tendon underneath. Place and label free nerve endings, a tactile disc, a hair follicle receptor, a tactile corpuscle, a lamellar corpuscle, a bulbous corpuscle, a muscle spindle, and a tendon organ where the notes put them. Write what each one detects.

Then: Beside the drawing, write the three ways receptors are classified, and tag each receptor you drew as an exteroceptor or a proprioceptor.

B RECEPTOR TABLE

Draw a table with the nine named receptors down the side, from free nerve endings to joint kinesthetic receptors. Across the top: where it sits, what it detects, and its class by stimulus (such as mechanoreceptor or nociceptor). Use a word or a tiny sketch in each cell.

Then: Below the table, list the three classes by location and the two by structure, and write one receptor from your table beside each class where one fits.

The Cranial Nerves

190

Cranial nerves

MODULE 5 · LECTURE · LAB

You should be able to: Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen.

A LABELED INFERIOR VIEW

Draw the inferior view of the brain and brainstem. Draw and label all twelve cranial nerves, I to XII, where each one attaches, along with the olfactory bulb and tract, optic chiasm, optic tract, and pontomedullary junction. Beside each nerve, write S, M, or B for sensory, motor, or both.

Then: Write the skull foramen at the end of each nerve, and circle the three openings that carry more than one nerve.

B CRANIAL NERVE TABLE

Draw a table with CN I to CN XII down the side. Across the top: name, class (sensory, motor, or mixed), primary function, and skull foramen. Write the names in order from memory first, using the memory sentence from the notes, then fill the other columns. Use a word or a tiny sketch in each cell.

Then: Color the rows that pass through the superior orbital fissure, the internal acoustic meatus, and the jugular foramen, using one color per opening.

MODULE 5

The Nerve Plexuses

191

Nerve plexuses

MODULE 5 · LECTURE · LAB

You should be able to: Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged.

A LABELED PLEXUS MAP

Draw the spinal cord with the spinal nerves leaving one side, grouped and numbered by region. Bracket and label the cervical, brachial, lumbar, sacral, and coccygeal plexuses, and write the spinal nerves that form each one. Draw the intercostal nerves running straight out from the thoracic region without forming a plexus.

Then: On a small body outline beside it, shade in matching colors the region that loses movement and sensation when each plexus is damaged.

B DAMAGE TABLE

Draw a table with the five plexuses down the side. Across the top: the spinal nerves that form it, the region of the body it serves, and what a patient loses if it is damaged. Use a word or a tiny sketch in each cell.

Then: Below the table, draw the phrenic nerve leaving its plexus to reach the diaphragm, and write what happens to breathing if it is damaged.

MODULE 5

The Autonomic Nervous System

192

Autonomic two-neuron pathway

MODULE 5 · LECTURE

You should be able to: Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector.

A SIDE BY SIDE PATHWAYS

Draw a somatic pathway and an autonomic pathway side by side, each starting in the spinal cord. Show the somatic one as a single neuron running to skeletal muscle. Show the autonomic one as a preganglionic neuron, an autonomic ganglion, and a postganglionic neuron reaching an effector. Label the cell bodies, the myelinated and unmyelinated axons, and the synapse.

Then: Write the three autonomic effectors at the end of the autonomic pathway, and write voluntary or involuntary under each pathway.

B COMPARE TABLE

Draw a table with five features down the side: effector, control, number of motor neurons, effect on the effector, and awareness. Across the top: somatic nervous system and autonomic nervous system. Use a word or a tiny sketch in each cell.

Then: Below the table, draw the four-step autonomic route as a chain of boxes with arrows, from the preganglionic neuron to the effector tissue.

193

Autonomic ganglia and divisions

MODULE 5 LECTURE

You should be able to: Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses.

A LABELED GANGLIA MAP

Draw the brainstem and spinal cord down the middle with the sympathetic trunk beside the vertebral column. Bracket where each division leaves the CNS. Draw and label a sympathetic trunk ganglion, the prevertebral ganglia near the large abdominal arteries, and a terminal ganglion in an organ wall. Add the cardiac plexus on the heart, the celiac plexus on the abdominal organs, and the hypogastric plexus on the pelvic organs.

Then: Draw a small piece of gut wall and place the myenteric and submucosal plexuses between the layers where each one sits.

B COMPARE TABLE

Draw a table with sympathetic and parasympathetic across the top. Down the side: other name, origin in the CNS, location of its ganglia, the named ganglia that belong to it, and overall role. Use a word or a tiny sketch in each cell.

Then: Below the table, list the autonomic plexuses with what each supplies, and the two enteric plexuses with the gut layers each sits between.

PART B

The note sheets.

One sheet per competency, 61 sheets. The competency and both prompts sit on the left so nothing else has to be printed beside it. The box on the right is the whole assignment.

The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

133

COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Kidney gross anatomy and layers

You should be able to: Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia.

PICK ONE AND DO THAT ONE

☐ **A** LABELED LOCATION VIEW

Draw the back wall of the abdomen with the vertebrae from T12 to L3, and place both kidneys with the right one slightly lower. On one kidney, label the renal hilum with the ureter, renal artery, renal vein, and nerves. Around the other, draw and label the renal capsule, adipose capsule, and renal fascia.

Then: Write beside the right kidney why it sits lower, and write beside the left what retroperitoneal tells you about where it sits.

☐ **B** LAYERS TABLE

Draw a table with the renal capsule, adipose capsule, renal fascia, and renal hilum down the side. Across the top: where it sits (inner, middle, outer, or medial border), what it is made of or what passes through it, and its job. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch both kidneys between T12 and L3 with the right one lower, and mark the hilum on each.

DRAW IT

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PASS 2 COLOR _____

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Dr. Sharilyn Rennie

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134
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Internal kidney regions

You should be able to: Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FRONTAL SECTION

Draw a kidney cut in a frontal section. Label the renal capsule, renal cortex, renal medulla, the renal pyramids with their bases toward the cortex, the renal columns between them, the renal papillae, minor calyces, major calyces, renal pelvis, and renal sinus. Show the ureter leaving at the hilum.

Then: Shade the parenchyma in one color and write beside it the two regions it includes and what it holds.

☐ **B** SORTING BOX

Draw two boxes, one labeled parenchyma and one labeled spaces that collect urine. Sort the cortex, renal columns, renal pyramids, renal papillae, minor calyces, major calyces, and renal pelvis into the two boxes. Beside the boxes, draw one pyramid with its papilla draining into a minor calyx, then a major calyx and the renal pelvis.

Then: Label a renal column beside your pyramid and write which region of the kidney it is an extension of.

DRAW IT

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135
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Nephron structure

You should be able to: Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron.

PICK ONE AND DO THAT ONE

☐ **A** LABELED NEPHRON

Draw one nephron with its collecting duct. Label the renal corpuscle, the glomerulus, the glomerular capsule with its parietal and visceral layers, the podocytes, the capsular space, the proximal convoluted tubule, the descending and ascending limbs of the nephron loop, the distal convoluted tubule, and the collecting duct.

Then: Draw arrows for the filtrate from the capsular space to the collecting duct, and mark the segment lined with a brush border.

☐ **B** PARTS TABLE

Draw a table with the glomerulus, glomerular capsule, podocytes, capsular space, proximal convoluted tubule, nephron loop, distal convoluted tubule, and collecting duct down the side, with the renal corpuscle parts first and then the tubule parts in the order the filtrate meets them. Across the top: part of the renal corpuscle or the tubule, and one feature you would use to spot it. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a podocyte wrapped around a glomerular capillary and label its foot processes and the filtration slits.

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136

COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Cortical vs juxtamedullary nephrons

You should be able to: Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds.

PICK ONE AND DO THAT ONE

☐ **A** SIDE BY SIDE NEPHRONS

Draw a strip of kidney with the cortex on top, the medulla below, and a line between them. Draw a cortical nephron and a juxtamedullary nephron side by side, showing how deep each nephron loop reaches. Wrap the cortical nephron in peritubular capillaries and the long juxtamedullary loop in the vasa recta.

Then: Write the percent of all nephrons each type makes up beside it, and star the type that concentrates the urine.

☐ **B** COMPARE TABLE

Draw a table with cortical nephrons and juxtamedullary nephrons down the side. Across the top: share of all nephrons, length of the loop, how far the loop reaches into the medulla, and the capillary bed that serves it. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch one renal pyramid with a short loop and a long loop drawn into it, and label each loop's nephron type.

DRAW IT

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137
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

JGA and filtration membrane

You should be able to: Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CORPUSCLE CLOSE-UP

Draw a renal corpuscle with its afferent and efferent arterioles and the distal convoluted tubule touching them. Label the juxtaglomerular apparatus, the macula densa cells in the tubule, and the juxtaglomerular (granular) cells in the afferent arteriole. Add a zoomed box of the filtration membrane and label the fenestrated endothelium, basement membrane, and slit membranes.

Then: Draw a blood cell, a large protein, and a medium protein, each stopped at the layer that holds it back.

☐ **B** WHERE AND WHAT TABLE

Draw a table with the macula densa cells, the juxtaglomerular cells, and the three layers of the filtration membrane down the side, the layers in order from the blood to the capsular space. Across the top: where it sits, and what it senses, releases, or holds back. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the distal convoluted tubule touching the arterioles of its own corpuscle and circle where the juxtaglomerular apparatus forms.

DRAW IT

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138
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Kidney blood supply pathway

You should be able to: Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VESSEL CUTAWAY

Draw a frontal section of a kidney showing two pyramids and the cortex. Draw and label the renal artery at the hilum, the segmental arteries branching in the renal sinus, the interlobar arteries running between the pyramids, the arcuate arteries along the pyramid bases, and the interlobular (cortical radiate) arteries up into the cortex. Add one nephron with its afferent arteriole, glomerulus, efferent arteriole, and peritubular capillaries.

Then: Draw the veins in a second shade beside the arteries and label the cortical radiate vein, arcuate vein, and renal vein.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows: renal artery, segmental and interlobar arteries, arcuate artery, interlobular (cortical radiate) artery, afferent arteriole, glomerulus, efferent arteriole, peritubular capillaries and vasa recta, cortical radiate vein, arcuate vein, interlobar vein, renal vein. Under each artery box, write a word or two saying where it runs.

Then: Circle the two capillary beds in your chain and box the one vessel that sits between them.

DRAW IT

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139

COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Ureters, bladder, urethra and urine path

You should be able to: Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra.

PICK ONE AND DO THAT ONE

☐ **A** LABELED URINARY TRACT

Draw a kidney, ureter, and bladder, with the kidney and bladder cut open. Label the minor and major calyces, renal pelvis, ureter, detrusor muscle in the bladder wall, the trigone with the two ureteral openings and the internal urethral orifice, the internal urethral sphincter, and the external urethral sphincter. Add arrows for urine from the collecting duct to the outside.

Then: Beside the bladder, sketch a male and a female urethra, and label the prostatic, membranous, and spongy parts of the male one.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows for urine: collecting duct, renal papilla, minor calyx, major calyx, renal pelvis, ureter, urinary bladder, urethra. Under the bladder box, list the trigone, detrusor muscle, and both urethral sphincters. Split the urethra box into male (prostatic, membranous, spongy) and female, and write which one is longer.

Then: Write beside each sphincter whether it is smooth or skeletal muscle and whether you control it.

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140
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE

Clinical anatomy of urinary disorders

You should be able to: Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw the urinary tract from the kidneys to the urethra, with the adipose capsule around one kidney. Write each disorder on the structure it affects: nephroptosis, kidney stones, urinary tract infection, glomerulonephritis, renal failure, and polycystic kidney disease. Draw one stone in the renal pelvis and another lodged in the ureter.

Then: Draw a dropped kidney beside your map with a kinked ureter, and write what loss of fat has to do with it.

☐ **B** DISORDER TABLE

Draw a table with nephroptosis, kidney stones, urinary tract infection, glomerulonephritis, and polycystic kidney disease down the side. Across the top: the structure it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Below the table, draw a short female urethra beside a long male urethra and write why urinary tract infections are more common in women.

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141 COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Scrotum and testis structure

You should be able to: Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules, sustentacular and interstitial cells, rete testis, and efferent ductules.

PICK ONE AND DO THAT ONE

☐ **A** LABELED TESTIS SECTION

Draw the scrotum with one testis cut open inside it. Label the scrotal septum, dartos muscle, cremaster muscle, tunica vaginalis, tunica albuginea, lobules, and seminiferous tubules. Add a zoomed circle of one tubule cut across, and label the sustentacular cells in its wall and the interstitial cells between the tubules.

Then: Write beside the dartos and the cremaster which kind of muscle each one is and what it moves.

☐ **B** STRUCTURE TABLE

Draw a table with the scrotum, dartos muscle, cremaster muscle, tunica vaginalis, tunica albuginea, seminiferous tubules, sustentacular cells, and interstitial cells down the side. Across the top: where it sits (scrotum, covering the testis, or inside the testis) and its job. Use a word or a tiny sketch in each cell.

Then: Circle the row whose cells produce testosterone, and star the two rows that set the temperature of the testes.

DRAW IT

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142

COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Sperm duct system and pathway

You should be able to: Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents.

PICK ONE AND DO THAT ONE

☐ **A** LABELED DUCT VIEW

Draw a side view of the male pelvis with the testis, the epididymis on it, and the ductus deferens running up out of the scrotum in the spermatic cord. Label the seminal vesicle, the ejaculatory duct, the prostate, and the prostatic, membranous, and spongy urethra. Add arrows for sperm from the seminiferous tubules to the outside.

Then: Draw the spermatic cord cut across as a circle, and label the ductus deferens, blood vessels, nerves, and cremaster muscle inside it.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows: seminiferous tubules, epididymis, ductus deferens, ejaculatory duct, prostatic urethra, membranous urethra, spongy urethra. Under each box, write in a word or two what happens to the sperm there. Box the one structure that produces sperm.

Then: Draw a bracket over the duct that runs inside the spermatic cord, and list the cord's other three contents beside it.

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DRAW IT

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143 COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Sperm cell structure

You should be able to: Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SPERM CELL

Draw one sperm cell large enough to fill the box. Label the head with its nucleus, the acrosome capping the head, the neck, the middle piece, the principal piece, and the end piece. Draw the middle piece packed with small mitochondria and the end piece tapering to a point.

Then: Write beside each part what it holds or does, in three words or fewer.

☐ **B** PARTS TABLE

Draw a table with the head, acrosome, neck, middle piece, principal piece, and end piece down the side, in order from front to back. Across the top: how you spot it and what it holds or does. Use a word or a tiny sketch in each cell.

Then: Draw a bracket beside the rows that make up the tail, and star the longest part of the tail.

DRAW IT

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144

COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Accessory glands and penis

You should be able to: Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis.

PICK ONE AND DO THAT ONE

☐ **A** LABELED PENIS VIEW

Draw a side view of the penis and a cross section of its body. On the side view, label the root, body, glans penis, and prepuce. On the cross section, label the two corpora cavernosa on the dorsal side and the corpus spongiosum on the ventral side surrounding the spongy urethra.

Then: Add the seminal vesicles, prostate, and bulbourethral glands as labeled boxes beside your drawing, and write what fluid each adds to semen.

☐ **B** TWO PART TABLE

Draw a table in two parts. First, put the seminal vesicles, prostate, and bulbourethral glands down the side, with what each adds to semen across the top. Second, put the root, body, glans penis, prepuce, corpora cavernosa, and corpus spongiosum down the side, with what each is. Use a word or a tiny sketch in each cell.

Then: Star the gland that adds the largest share of semen, and circle the erectile column that forms the glans.

DRAW IT

split it up however the prompt needs

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145
COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE

Clinical anatomy of male disorders

You should be able to: Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw a side view of the male pelvis with the testis in the scrotum, the inguinal canal, and the urethra passing through the prostate. Write each disorder on the structure it affects: cryptorchidism, hydrocele, varicocele, benign prostatic hyperplasia, prostate cancer, testicular cancer, and inguinal hernia.

Then: Draw an undescended testis with a dotted line to where it should sit, and write why staying at body temperature harms it.

☐ **B** DISORDER TABLE

Draw a table with cryptorchidism, hydrocele, varicocele, benign prostatic hyperplasia, and inguinal hernia down the side. Across the top: the structure it affects, where that structure sits, and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch an enlarged prostate squeezing the urethra and write the urinary symptom it causes.

DRAW IT

split it up however the prompt needs

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146

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Ovary structure and follicle stages

You should be able to: Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans.

PICK ONE AND DO THAT ONE

☐ **A** LABELED OVARY VIEW

Draw the uterus with one ovary beside it. Label the ovarian ligament, suspensory ligament, broad ligament, and mesovarium. Draw the ovary cut open and label the germinal epithelium, tunica albuginea, cortex, and medulla. In the cortex, draw ovarian follicles, a mature follicle, a corpus luteum, and a corpus albicans.

Then: Number the follicle structures in the order they appear, and mark where the oocyte leaves at ovulation.

☐ **B** SEQUENCE STRIP

Draw a row of circles showing a follicle growing into a mature follicle, releasing its oocyte, then becoming a corpus luteum and finally a corpus albicans. Beneath the row, list the ovary layers from the outside in, then the three ligaments with what each one attaches the ovary to.

Then: Circle the layer of the ovary where every follicle in your strip is found.

DRAW IT

split it up however the prompt needs

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Dr. Sharilyn Rennie

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147

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Uterine tubes and ovum pathway

You should be able to: Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED TUBE VIEW

Draw the uterus with one uterine tube and the ovary beside it. Label the fimbriae, infundibulum, ampulla, and isthmus, and show the infundibulum open to the pelvic cavity near the ovary. Draw an oocyte leaving a mature follicle, with arrows following it through the tube into the uterine cavity.

Then: Star where fertilization usually happens, and draw cilia on the tube lining with an arrow showing which way they move the oocyte.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows: ovary, fimbriae and infundibulum, ampulla, isthmus, uterus. Under each box, write what happens to the oocyte there and one feature of that region, such as its shape or width. Use a word or a tiny sketch under each box.

Then: Draw a dotted arrow from the open infundibulum into the pelvic cavity, and write one disorder this open route explains.

DRAW IT

split it up however the prompt needs

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Dr. Sharilyn Rennie

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148

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Uterus regions, ligaments, and wall

You should be able to: Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium.

PICK ONE AND DO THAT ONE

☐ **A** LABELED UTERUS VIEW

Draw the uterus cut open from the front, with the uterine tubes and the top of the vagina. Label the fundus, body, cervix, uterine cavity, and cervical canal with the internal and external os. Label the perimetrium, myometrium, and endometrium in the wall. Add the broad, round, uterosacral, and cardinal ligaments as lines running to what each attaches to.

Then: Zoom in on the endometrium and split it into the layer that sheds each month and the layer that rebuilds it.

☐ **B** REGION STACK AND TABLE

Draw a stack of three boxes from top to bottom for the fundus, body, and cervix, with the uterine cavity and cervical canal drawn inside. Beside it, draw a table with the four ligaments and the three wall layers down the side. Across the top: what it connects to or what it is made of. Use a word or a tiny sketch in each cell.

Then: Star the wall layer that contracts during childbirth, and circle the ligament that runs to the labia majora.

DRAW IT

split it up however the prompt needs

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Dr. Sharilyn Rennie

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149

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Vagina and external genitalia

You should be able to: Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VULVA VIEW

Draw the vulva in an inferior view, with the mons pubis at the top. Label the labia majora, labia minora, clitoris, vestibule, urethral orifice, vaginal orifice, hymen, and the greater vestibular glands beside the vaginal opening. Outline the perineum around the genitalia and the anus.

Then: Sketch a side view of the vagina meeting the cervix, and label the fornix around the cervix.

☐ **B** STRUCTURE TABLE

Draw a table with the vagina, fornix, hymen, mons pubis, labia majora, labia minora, clitoris, vestibule, and greater vestibular glands down the side. Across the top: where it sits and what it is or does. Use a word or a tiny sketch in each cell.

Then: Mark the two structures that are homologous to male structures, and write the male match beside each.

DRAW IT

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150

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Mammary gland structure

You should be able to: Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BREAST SECTION

Draw a side view of a breast cut open, sitting over the chest muscles. Label two or three lobes, the lobules inside one lobe, the alveoli inside one lobule, the lactiferous ducts running to the nipple, the areola, and the suspensory ligaments.

Then: Draw arrows for the path of milk from the alveoli to the nipple, and label each structure the arrows pass.

☐ **B** NESTING DIAGRAM

Draw nested shapes: alveoli inside a lobule, lobules inside a lobe, and the lobe draining by a lactiferous duct to the nipple. Beside it, draw a small table with the nipple, areola, and suspensory ligaments down the side and what each is or does across the top. Use a word or a tiny sketch in each cell.

Then: Write beside your lobe how many lobes one mammary gland has.

DRAW IT

split it up however the prompt needs

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151

COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Brain gross organization

You should be able to: Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MIDSAGITTAL VIEW

Draw the brain cut down the middle and seen from the side. Label the cerebrum, diencephalon, brainstem, and cerebellum, and outline each one so the four divisions are easy to tell apart. Along the edge of the cerebrum, draw a thin outer band of gray matter over a thick core of white matter, and label both.

Then: Write beside the gray matter band what it is made of, and beside the white matter what it is made of.

☐ **B** DIVISION TABLE

Draw a table with the cerebrum, diencephalon, brainstem, and cerebellum down the side. Across the top: where it sits and its main job. Use a word or a tiny sketch in each cell. Below the table, draw a small block of brain tissue with gray matter on the outside and white matter on the inside, and label both.

Then: Star the largest division and the second largest division, and write the cranial fossa where the cerebellum sits.

DRAW IT

split it up however the prompt needs

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152
COMP

MODULE 5 · THE BRAIN · LECTURE

Neural tube brain development

You should be able to: Trace brain development from the neural tube through the primary and secondary vesicles and match each secondary vesicle to its adult structures and ventricle.

PICK ONE AND DO THAT ONE

☐ **A** BRANCHING FLOW CHAIN

Draw the neural tube at the top of the box. Branch it into the three primary vesicles, writing each one's formal name and common name (forebrain, midbrain, hindbrain). Branch those into the five secondary vesicles. Under each secondary vesicle, write the adult structures it becomes and its ventricle or fluid space.

Then: Write the germ layer the neural tube forms from, and circle the one primary vesicle that carries through without splitting.

☐ **B** VESICLE TABLE

Write neural tube at the top of the box. Below it, draw a table with the five secondary vesicles down the side. Across the top: the primary vesicle it came from, the adult structures it becomes, and its ventricle or fluid space. Use a word or a tiny sketch in each cell.

Then: Draw a simple side view of the adult brain and number each region 1 to 5 to match the secondary vesicle it grew from.

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153
COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Cerebral surface and lobes

You should be able to: Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen.

PICK ONE AND DO THAT ONE

☐ **A** SIDE AND TOP VIEWS

Draw the left side of the brain. Label the frontal, parietal, temporal, and occipital lobes, a gyrus, a sulcus, the precentral and postcentral gyri, and the central, lateral, and parieto-occipital sulci. Draw the insula as a dashed shape inside the lateral sulcus. Show the transverse fissure above the cerebellum. Beside it, draw a small top view and label both hemispheres and the longitudinal fissure.

Then: Write beside each of the four outer lobes the cranial bone that lies over it and gives it its name.

☐ **B** BOUNDARY TABLE

Sketch three folds of cortex and label a gyrus, a sulcus, and a fissure. Below, draw a table with the central sulcus, lateral sulcus, parieto-occipital sulcus, longitudinal fissure, and transverse fissure down the side. Across the top: the region on one side of it and the region on the other. Use a word or a tiny sketch in each cell.

Then: Draw the brain seen from above, label the longitudinal fissure, central sulcus, and each lobe you can see, then name the lobe no outside view shows.

DRAW IT

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154
COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Functional cortical areas

You should be able to: Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CORTEX MAP

Draw the side view of the hemisphere where Broca's and Wernicke's areas are usually found. Draw the central and lateral sulci. Label the primary motor cortex, primary somatosensory cortex, Broca's area, Wernicke's area, prefrontal cortex, primary auditory cortex, and primary visual cortex. Draw the insula dashed for the gustatory cortex, and point an arrow to the medial temporal lobe for the olfactory cortex.

Then: Write M, S, or A on each labeled area to sort it as a motor, sensory, or association area.

☐ **B** AREA LOCATION TABLE

Draw a table with these areas down the side: primary motor, primary somatosensory, primary visual, primary auditory, primary gustatory, and primary olfactory cortex, Broca's area, Wernicke's area, and the prefrontal cortex. Across the top: lobe, gyrus or landmark, motor, sensory, or association, and job. Use a word or a tiny sketch in each cell.

Then: Add a row for each association area the notes place next to a primary area, and name the primary area it sits beside.

DRAW IT

split it up however the prompt needs

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155

COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Cerebral white matter tracts

You should be able to: Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FRONTAL SECTION

Draw a frontal section through both hemispheres with gray cortex on the outside and white matter inside. Draw and label an association fiber arching between two gyri on the same side, the corpus callosum crossing between the hemispheres, and the projection fibers of the internal capsule running down beside the basal ganglia toward the brainstem.

Then: Shade the internal capsule and write beside it why a small stroke there can weaken a whole side of the body.

☐ **B** FIBER CLASS TABLE

Draw a table with association, commissural, and projection fibers down the side. Across the top: what it connects, a named example, and a tiny arrow sketch of its path. Use a word or a tiny sketch in each cell. Below, sketch a midline view with the corpus callosum and a frontal section with the internal capsule, and label both.

Then: Circle the largest commissural tract, and write which tract links Broca's and Wernicke's areas and which fiber class it belongs to.

DRAW IT

split it up however the prompt needs

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COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Basal ganglia

You should be able to: Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FRONTAL SECTION

Draw a frontal section of one hemisphere with cortex on the outside, white matter inside, and deep masses of gray matter buried in the white matter. Label the caudate nucleus, the putamen, and the globus pallidus medial to the putamen. Below the hemisphere, sketch the diencephalon and midbrain, and mark the subthalamic nucleus and the substantia nigra.

Then: Draw dotted lines linking all five nuclei, and write the three roles the basal ganglia share.

☐ **B** LOCATION SORT

Draw three boxes stacked from top to bottom labeled cerebrum, diencephalon, and midbrain. Write each nucleus in the box where it sits: caudate nucleus, putamen, globus pallidus, subthalamic nucleus, and substantia nigra. Circle the three core nuclei and underline the two grouped with them only by function. Add an arrow showing the putamen lateral to the globus pallidus.

Then: Write beside the substantia nigra what it looks like and the disease its loss causes, and give one semi-voluntary movement the basal ganglia control.

DRAW IT

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COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Diencephalon

You should be able to: Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MIDSAGITTAL VIEW

Draw the brain cut down the midline. Label the thalamus with the massa intermedia, the hypothalamus below it, the epithalamus at the back with the pineal gland, and the third ventricle. Add the corpus callosum above, the infundibulum and optic chiasm below the hypothalamus, and the midbrain below as landmarks.

Then: Shade the third ventricle, and write the percent of the diencephalon the thalamus makes up.

☐ **B** THREE PART TABLE

Draw a table with the thalamus, hypothalamus, and epithalamus down the side. Across the top: where it sits, what it contains, and its main job. Use a word or a tiny sketch in each cell. Below the table, draw the third ventricle as a midline cavity and label it.

Then: Draw the right and left thalami on each side of your third ventricle, and join them across it with the massa intermedia.

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COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Cerebellum

You should be able to: Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections.

PICK ONE AND DO THAT ONE

☐ **A** BACK AND CUT VIEWS

Draw the cerebellum from behind and label the two hemispheres, the vermis, and the folia. Beside it, draw the cerebellum cut down the midline and attached to the brainstem. Label the cerebellar cortex, arbor vitae, a deep cerebellar nucleus, and the superior, middle, and inferior cerebellar peduncles as stalks running to the midbrain, pons, and medulla.

Then: Shade the gray matter on your cut view and leave the white matter blank.

☐ **B** PARTS TABLE

Draw a table with these down the side: cerebellar hemispheres, vermis, folia, arbor vitae, deep cerebellar nuclei, and the superior, middle, and inferior cerebellar peduncles. Across the top: what it is, gray or white matter, and where it sits or what it joins. Use a word or a tiny sketch in each cell.

Then: Draw an arrow on each peduncle row showing whether it mostly carries signals into the cerebellum or out of it.

DRAW IT

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COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Limbic system

You should be able to: Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MIDSAGITTAL VIEW

Draw the brain cut down the midline. Label the corpus callosum, the cingulate gyrus arching above it, and the diencephalon beneath it. Draw the hippocampus and amygdala as dashed shapes curving down and forward below the diencephalon, so that the three limbic structures form a ring around it.

Then: Write beside each limbic structure its main role from the notes.

☐ **B** LIMBIC COMPARE TABLE

Draw a table with the cingulate gyrus, amygdala, and hippocampus down the side. Across the top: where it sits, its role, and a tiny sketch. Use a word or a tiny sketch in each cell. Below the table, draw a small oval for the diencephalon and arrange the three structures around it as a ring.

Then: Write in the center of your ring the three jobs of the limbic system as a whole.

DRAW IT

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160
COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Cerebral arterial supply

You should be able to: Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches.

PICK ONE AND DO THAT ONE

☐ **A** LABELED INFERIOR VIEW

Draw the base of the brain seen from below with the brainstem in the middle. Draw the vertebral arteries joining into the basilar artery, then the ring: posterior cerebral, posterior communicating, internal carotid, anterior cerebral, and anterior communicating arteries. Add the middle cerebral arteries and the cerebellar branches: posterior inferior, anterior inferior, and superior cerebellar arteries.

Then: Shade the anterior circulation one way and the posterior circulation another, and write what each one feeds.

☐ **B** TWO CIRCULATION CHAINS

Draw two flow chains side by side. Anterior: internal carotid artery to the anterior and middle cerebral arteries. Posterior: vertebral arteries to basilar artery to posterior cerebral arteries, with the posterior inferior cerebellar branch off each vertebral and the anterior inferior and superior cerebellar branches off the basilar. Join the two chains with the posterior communicating artery, and add the anterior communicating artery joining the right and left anterior cerebral arteries.

Then: Write beside each cerebral artery the part of the brain it supplies.

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Brainstem regions

You should be able to: Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SIDE VIEW

Draw the brainstem from the side with the diencephalon above it, the cerebellum behind it, and the spinal cord below. Label the midbrain, pons, and medulla oblongata from top to bottom, the basilar pons, and the pontomedullary junction. Mark the foramen magnum where the medulla becomes the spinal cord.

Then: Draw one arrow up and one arrow down through all three regions, and label them ascending sensory tracts and descending motor tracts.

☐ **B** REGION ORDER CHAIN

Draw a vertical chain of boxes from top to bottom: cerebrum and diencephalon, midbrain, pons, medulla oblongata, and spinal cord, with the cerebellum to the side. In each brainstem box, write its position (most superior, middle, most inferior) and one structure you can find on it on the model.

Then: Write beside the chain what passes through all three regions and how many cranial nerves have nuclei in the brainstem.

DRAW IT

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Midbrain structures

You should be able to: Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen.

PICK ONE AND DO THAT ONE

☐ **A** TWO LABELED VIEWS

Draw the midbrain from behind and in cross section. On the back view, label the tectum and its four colliculi, with the superior pair above the inferior pair. On the cross section, label the tectum, the cerebral aqueduct, the darkly pigmented substantia nigra, and the cerebral peduncles on the front.

Then: Draw an arrow into and out of the cerebral aqueduct, and name the ventricle at each end.

☐ **B** MIDBRAIN STRUCTURE TABLE

Draw a table with the tectum, superior colliculi, inferior colliculi, cerebral peduncles, substantia nigra, and cerebral aqueduct down the side. Across the top: how to spot it on the model, and its job. Use a word or a tiny sketch in each cell.

Then: Circle the two reflex centers, and write beside them whether you become conscious of the sight or sound there.

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Pons structures

You should be able to: Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SIDE VIEW

Draw the brainstem from the left side with the cerebellum behind it. Label the midbrain, pons, and medulla, the basilar pons, and the thick middle cerebellar peduncle running back into the cerebellum. Dot the pontine nuclei inside the pons. Draw the fourth ventricle as a dashed space behind the pons and upper medulla.

Then: Draw an arrow from the cerebral cortex through the pontine nuclei and the middle cerebellar peduncle into the cerebellum.

☐ **B** PONS FLOW CHAIN

Draw a flow chain for motor information: cerebral cortex, pontine nuclei, middle cerebellar peduncles, cerebellum. Write what each step does in a few words. Under the chain, sketch the pons with the midbrain above, the medulla below, and the fourth ventricle behind it, and label all four.

Then: Write what the word pons means, and name the three regions it bridges.

DRAW IT

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Medulla oblongata structures

You should be able to: Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ANTERIOR VIEW

Draw the brainstem from the front with the spinal cord below. Label the pons, the pontomedullary junction, the medulla oblongata, the two pyramids, the medullary olive on each side of the pyramids, and the decussation of the pyramids where the medulla meets the cord. Inside the medulla, draw and label boxes for the cardiac, respiratory, and vasomotor centers.

Then: Draw crossing arrows at the decussation, and write why a stroke in the right hemisphere weakens the left side of the body.

☐ **B** MEDULLA TABLE

Draw a table with the pyramids, medullary olive, decussation of the pyramids, cardiac center, respiratory center, and vasomotor center down the side. Across the top: seen on the surface or inside the medulla, and what it looks like or what it does. Use a word or a tiny sketch in each cell.

Then: Star the three vital centers, and write why injury to the medulla is immediately life threatening.

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Cranial nerve nuclei distribution

You should be able to: Match cranial nerve nuclei III through XII to the brainstem region that houses them.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BRAINSTEM VIEW

Draw the brainstem from the front, divided into the midbrain, pons, and medulla oblongata, and label the pontomedullary junction. Beside each region, write the Roman numerals and names of the cranial nerves whose nuclei it holds, from CN III down to CN XII.

Then: Write which two cranial nerves have no nuclei in the brainstem, and circle the one nerve you wrote beside two regions.

☐ **B** NERVE SORT TABLE

Draw a table with CN III through CN XII down the side. Across the top: the nerve's name, midbrain, pons, and medulla oblongata. Write each name, then put a check in the region that houses that nerve's nuclei. Use a word or a check in each cell.

Then: Count how many of the twelve cranial nerves have nuclei in the brainstem, and circle the nerve with checks in two regions.

DRAW IT

split it up however the prompt needs

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166

COMP

MODULE 5 · THE BRAINSTEM · LECTURE

Reticular formation

You should be able to: Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CUTAWAY

Draw the brainstem from the side, cut open to show its core. Draw the reticular formation as a net of dots and lines woven through the midbrain, pons, and medulla. Draw arrows up from the net to the cerebral cortex labeled reticular activating system, and arrows down toward the spinal cord labeled descending component.

Then: Write beside each set of arrows two of its jobs from the notes.

☐ **B** COMPARE TABLE

Draw a table with the reticular activating system and the descending component down the side. Across the top: which way it sends signals, what it acts on, and its jobs. Use a word or a tiny sketch in each cell. Above the table, sketch the brainstem core with a gray matter net running through its white matter.

Then: Write under the table what happens when the reticular activating system shuts down and what severe damage to it can cause.

DRAW IT

split it up however the prompt needs

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167

COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Cranial meninges and dural reflections

You should be able to: Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MIDSAGITTAL VIEW

Draw the inside of the skull in midsagittal view with the brain removed. Label the falx cerebri, falx cerebelli, and tentorium cerebelli. Draw the superior sagittal sinus on the falx cerebri, then the confluence of sinuses, a transverse sinus, a sigmoid sinus, and the internal jugular vein. In a corner, draw the dura, arachnoid, and pia from skull to brain.

Then: Add arrows along the sinuses for the direction blood drains, and label the periosteal and meningeal layers on your dura strip.

☐ **B** TABLE AND FLOW CHAIN

Draw a table with the dura mater, arachnoid mater, pia mater, falx cerebri, falx cerebelli, and tentorium cerebelli down the side. Across the top: where it sits and what it touches or separates. Use a word or a tiny sketch in each cell. Below the table, draw a chain of boxes from the superior sagittal sinus to the internal jugular vein.

Then: Mark the box in your chain where the arachnoid granulations return CSF to the blood.

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PASS 2 COLOR _____

DRAW IT

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Dr. Sharilyn Rennie

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Spinal meninges and meningeal spaces

You should be able to: Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CROSS SECTION

Draw a cross section of a vertebra with the spinal cord inside. Label the vertebral bone, the epidural space with fat and veins, the dura mater, subdural space, arachnoid mater, subarachnoid space with CSF, and pia mater. Draw a denticulate ligament from the pia to the dura. Beside it, sketch the lower cord with the filum terminale running to the coccyx.

Then: Star the space that holds CSF, and write potential space beside the space that is only a potential space.

☐ **B** SPACE COMPARE TABLE

Draw a table with the epidural, subdural, and subarachnoid spaces down the side. Across the top: the two layers it sits between, what fills it, and whether it is also found in the cranium. Use a word or a tiny sketch in each cell. Below the table, sketch the cord with two denticulate ligaments and the filum terminale.

Then: Write beside each anchor which meningeal layer it is made of and what it attaches the cord to.

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Cord termination and lumbar puncture

You should be able to: Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SIDE VIEW

Draw a side view of the lower vertebral column with the spinal cord inside. Label the conus medullaris at the L1 to L2 disc, the cauda equina, the filum terminale, and the lumbar cistern below the cord. Draw a needle entering in the midline between the L3 and L4 spines, and label each layer it crosses, in order, from the skin to the subarachnoid space.

Then: Write beside the conus medullaris why the needle is placed between the L3 and L4 or L4 and L5 spines.

☐ **B** NEEDLE PATH SEQUENCE

Draw a vertical stack of boxes, one for each layer a lumbar puncture needle crosses, in order from the skin at the top to the subarachnoid space at the bottom. Name the layer and sketch what it looks like in each box. Beside the bottom box, sketch the lumbar cistern with the floating cauda equina, the conus medullaris above it, and the filum terminale.

Then: Mark the box where a clinician feels a firm ligament give way, and write the two spine levels where the needle can enter.

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Ventricular system

You should be able to: Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VENTRICLE VIEW

Draw a side view of the brain outline with the ventricles inside it, as on the brain model. Label a lateral ventricle, an interventricular foramen of Monro, the third ventricle, the cerebral aqueduct, the fourth ventricle, and the central canal of the spinal cord. Draw the median aperture and the two lateral apertures opening from the fourth ventricle into the subarachnoid space.

Then: Write beside each ventricle the region of the brain it sits in, and write what lines every chamber.

☐ **B** CONNECTION TABLE

Draw a table with the lateral ventricles, third ventricle, fourth ventricle, and central canal down the side. Across the top: where it is, what it connects to next, and the opening or channel that makes the connection. Use a word or a tiny sketch in each cell.

Then: Circle the three narrow connection points in your table, and write beside the table what would happen if one were blocked.

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

CSF production and circulation

You should be able to: Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FLOW DRAWING

Draw a midsagittal view of the brain and upper spinal cord showing the ventricles, the subarachnoid space, and the superior sagittal sinus. Draw choroid plexus in the lateral, third, and fourth ventricles. Add numbered arrows for CSF from the lateral ventricles through the interventricular foramina, third ventricle, cerebral aqueduct, fourth ventricle, apertures, and subarachnoid space to the arachnoid granulations.

Then: Add a side arrow into the central canal, and write which cells of the choroid plexus produce the CSF.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows for the one-way loop of CSF, from production at the choroid plexus of the lateral ventricles to reabsorption at the arachnoid granulations. Give each ventricle, opening, and space its own box. Add a tiny sketch in each box of what the structure looks like.

Then: Mark each box that holds choroid plexus, and add a final arrow into the venous blood of the superior sagittal sinus.

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Blood–brain barrier structure

You should be able to: Name the three structural components of the blood-brain barrier the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CAPILLARY SECTION

Draw a cross section of a brain capillary with blood in its center. Label the capillary endothelial cells joined by tight junctions, the basement membrane wrapped around them, and the astrocyte end-feet covering the outside. Draw one whole star-shaped astrocyte with its end-feet reaching the capillary and a neuron nearby.

Then: Draw an arrow from the blood toward the neuron, and number each barrier component the arrow has to cross.

☐ **B** COMPARE TABLE

Draw a table with the three components of the blood-brain barrier down the side. Across the top: what it is, its position counting outward from the blood, and one feature that makes it part of the barrier. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch an ordinary capillary beside a brain capillary and label the pores on the ordinary capillary that the brain capillary lacks.

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COMP

MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE

Nervous system divisions

You should be able to: Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions.

PICK ONE AND DO THAT ONE

☐ **A** BRANCHING MAP

Draw a branching map that starts with the nervous system and splits into the CNS and PNS. Under the CNS, write the brain and spinal cord. Under the PNS, list nerves, ganglia, and sensory receptors, then branch into the somatic, autonomic, and enteric motor divisions. Split the autonomic division into sympathetic and parasympathetic.

Then: Write beside each motor division the effector it controls, and add the number of pairs of cranial and spinal nerves.

☐ **B** LABELED BODY OUTLINE

Draw a body outline with the brain and spinal cord shaded as the CNS. Draw a few nerves leaving the cord, one ganglion, and a sensory receptor in the skin, and label them as PNS. Draw a skeletal muscle, the heart, a gland, and a loop of gut wall with its own neurons.

Then: Label each effector with the motor division that controls it: somatic, autonomic, or enteric.

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COMP

MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Neuron structure

You should be able to: Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them.

PICK ONE AND DO THAT ONE

☐ **A** LABELED NEURON

Draw a multipolar neuron as it looks on the neuron model. Label the cell body with its nucleus and nucleolus, the Nissl bodies, the dendrites, the axon hillock, the axon, and the axon terminals with their synaptic bulbs.

Then: Add arrows along the neuron for the direction a signal travels, from the dendrites through to the axon terminals.

☐ **B** SIGNAL FLOW CHAIN

Draw a chain of boxes in the order a signal travels through a neuron: dendrites, cell body, axon hillock, axon, and axon terminals. In each box, sketch the part and write its job in a word or two. Draw Nissl bodies inside the cell body box and write what they are made of and what they do.

Then: Add neurofibrils to the cell body box, and write why an axon dies when it is cut away from its cell body.

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COMP

MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Synapse structures

You should be able to: Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SYNAPSE CLOSE-UP

Draw the end of an axon meeting a second neuron. Label the axon terminal, a synaptic end bulb, the synaptic vesicles inside it, the synaptic cleft, and the next cell. Draw neurotransmitter molecules leaving the vesicles and crossing the cleft. Beside it, draw three tiny sketches of an axon ending on a dendrite, on a cell body, and on another axon.

Then: Name each of your three tiny sketches by the site it contacts: axodendritic, axosomatic, or axoaxonic.

☐ **B** SYNAPSE TYPE MAP

Draw one neuron with three axons from other neurons ending on it: one on a dendrite, one on the cell body, and one on its axon. Name each synapse by the site it contacts. Draw a zoom circle on one synapse showing the synaptic end bulb, synaptic vesicles, and synaptic cleft.

Then: Write beside the zoom circle what the vesicles store and what carries the signal across the cleft.

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COMP

MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Neuron classification

You should be able to: Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each.

PICK ONE AND DO THAT ONE

☐ **A** THREE SHAPE SKETCHES

Draw a multipolar, a bipolar, and a unipolar neuron side by side. Label the dendrites, cell body, and axon on each one, and write how many processes leave the cell body. Under the multipolar and bipolar shapes, write where each is found, and add the Purkinje cell and pyramidal cell under the shape they belong to.

Then: Label which shape most sensory neurons have, and which shape motor neurons and interneurons have.

☐ **B** FUNCTIONAL PATHWAY MAP

Draw a skin receptor, a cross section of the spinal cord, and a skeletal muscle. Connect them with a sensory neuron carrying a signal toward the CNS, an interneuron inside the cord, and a motor neuron carrying a command out to the muscle. Draw each neuron in its structural shape and label the shape.

Then: In the margin, add a bipolar neuron and name the three places it is found.

DRAW IT

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MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Neuroglia identification

You should be able to: Name the four CNS glia and two PNS glia and describe the support role each one performs.

PICK ONE AND DO THAT ONE

☐ **A** LABELED GLIA SCENE

Draw a patch of CNS tissue with a neuron, a capillary, and the edge of a ventricle. Draw and label an astrocyte touching the capillary, an oligodendrocyte wrapping several axons, a microglial cell eating debris, and ependymal cells lining the ventricle. In a separate PNS box, draw Schwann cells on an axon and satellite cells around cell bodies in a ganglion.

Then: Write the support role of each glial cell beside its label in a few words.

☐ **B** SORTING BOX

Draw a box split into a CNS side and a PNS side. Sort the six glial cells into the correct side. For each cell, add a tiny sketch of its shape and a few words for the support role it performs.

Then: Circle the two cells that make myelin, and star the cell that helps form the blood-brain barrier.

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MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Myelination and regeneration

You should be able to: Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential.

PICK ONE AND DO THAT ONE

☐ **A** LABELED AXON COMPARISON

Draw a PNS axon wrapped by a row of Schwann cells. Label the myelin sheath, the nodes of Ranvier, and the neurolemma. Beside it, draw one CNS oligodendrocyte with arms wrapping segments of several different axons, and label its myelin sheath and nodes of Ranvier. Show that the CNS axons have no neurolemma.

Then: Draw a cut across each axon, then write which one can regrow and the reason beside it.

☐ **B** COMPARE TABLE

Draw a table with PNS and CNS across the top. Down the side: the cell that makes the myelin, how many axons one cell wraps, whether a neurolemma is present, and whether a cut axon can regrow. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a myelinated axon, write what myelin is made of, and draw the impulse jumping from node to node.

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COMP

MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE

CNS and PNS tissue terms

You should be able to: Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS.

PICK ONE AND DO THAT ONE

☐ **A** TWO BY TWO GRID

Draw a grid with CNS and PNS across the top. Down the side: a cluster of cell bodies and a bundle of axons. Write nucleus, ganglion, tract, and nerve in the correct boxes. Add a tiny sketch in each box, with dots for cell bodies and lines for axons.

Then: Write a one-line definition under each term that uses the words cluster or bundle and CNS or PNS.

☐ **B** LABELED NERVOUS SYSTEM MAP

Draw the brain and spinal cord with one nerve leaving the cord and a ganglion along that nerve outside the CNS. Draw a nucleus as a cluster of dots inside the brain and a tract as a bundle of lines running up the cord. Label all four structures.

Then: Draw a dashed line where the CNS ends and the PNS begins, and label each structure cell bodies or axons.

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MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE AND LAB

Gray and white matter

You should be able to: Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord.

PICK ONE AND DO THAT ONE

☐ **A** TWO LABELED SECTIONS

Draw a spinal cord cross section and a slice through the cerebrum side by side, as on the models. Shade gray matter one way and white matter another. On the brain, label the cortex and the white matter beneath it. On the cord, label the gray matter with its dorsal and ventral horns, the outer white matter, and the central canal.

Then: Add a small cerebellum with its gray cortex, and write what gray and white matter are made of in your shading key.

☐ **B** COMPARE TABLE

Draw a table with gray matter and white matter down the side. Across the top: what it is made of, where it sits in the brain, and where it sits in the spinal cord. Use a word or a tiny sketch in each cell.

Then: Below the table, draw two circles labeled brain and cord, and shade each to show which matter is on the outside.

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COMP

MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE

Reflex arc components

You should be able to: Name the five components of a reflex arc in order from receptor to effector and trace a signal through them.

PICK ONE AND DO THAT ONE

☐ **A** LABELED REFLEX ARC

Draw a receptor, a cross section of the spinal cord, and a muscle. Connect them with a sensory neuron running into the cord, a synapse in the cord, and a motor neuron running out to the muscle. Label the five components of the reflex arc and add arrows for the direction of the signal.

Then: Put an X at one spot on the arc and write what a missing reflex tells a clinician about that pathway.

☐ **B** FLOW CHAIN

Draw a chain of five boxes for the reflex arc in order, from receptor to effector. In each box, name the component, sketch it, and write its job in a few words. Draw an arrow between each pair of boxes for the direction of the signal.

Then: Circle the box where the synapses are, and write beside the chain why a reflex is fast.

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MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE

Types of reflexes

You should be able to: Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each.

PICK ONE AND DO THAT ONE

☐ **A** COMPARE TABLE

Draw a table with six reflex types down the side: somatic, autonomic, monosynaptic, polysynaptic, innate, and learned. Across the top: what the name is based on and one example. Use a word or a tiny sketch in each cell.

Then: Draw a bracket joining each pair of types, and write the one feature that splits that pair.

☐ **B** PAIRED SKETCHES

Draw three pairs of small sketches. For somatic and autonomic, show the effector each one uses. For monosynaptic and polysynaptic, show one synapse in the cord beside a chain with interneurons. For innate and learned, sketch one example of each. Label every sketch with its reflex type and its example.

Then: Write under each pair what the names are based on: the effector, the number of synapses, or whether it is inborn.

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MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE

Neuronal pools and pathways

You should be able to: Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CIRCUIT DRAWING

Draw a convergence circuit with many neurons feeding into one, and a divergence circuit with one neuron spreading to many, with arrows. Below them, draw a front view of the brain and spinal cord with one ascending and one descending pathway. Show each pathway crossing the midline and label the crossing decussation.

Then: Write beside the brain why a stroke on one side of the brain causes weakness on the opposite side.

☐ **B** COMPARE TABLE

Draw a table with convergence, divergence, ascending pathway, descending pathway, and decussation down the side. Across the top: what it is and which way the signal travels. Use a word or a tiny sketch in each cell.

Then: Below the table, draw a small body outline and shade the half the left side of the brain controls.

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COMP

MODULE 5 · THE SPINAL CORD · LECTURE AND LAB

Spinal cord external anatomy

You should be able to: Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED POSTERIOR VIEW

Draw the whole spinal cord from the back, running from the medulla down to its tapered end, with the vertebral column drawn faintly behind it. Label the cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, filum terminale, and posterior median sulcus. Mark the disc where the cord ends.

Then: Draw a small cross section beside it, label the anterior median fissure and posterior median sulcus, and show which groove is deeper.

☐ **B** STRUCTURE AND LANDMARK TABLE

Draw a table with seven structures down the side: cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus. Across the top: what it looks like, where it sits on the cord, and what it does or serves. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the lower cord and mark where a lumbar puncture needle goes, then write why that spot does not risk the cord.

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COMP

MODULE 5 · THE SPINAL CORD · LECTURE AND LAB

Roots, rootlets, and root ganglion

You should be able to: Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CUTAWAY

Draw a cross section of the spinal cord with a row of rootlets leaving the back and the front on one side. Gather them into the posterior root and the anterior root, add the posterior root ganglion, and join the two roots into a spinal nerve. Label each part and the intervertebral foramen where the nerve exits.

Then: Write sensory or motor beside each root, and add arrows showing signals coming in on one root and going out on the other.

☐ **B** FLOW CHAIN

Draw two chains of boxes with arrows. The sensory chain runs from a receptor in the skin through the spinal nerve, posterior root, and rootlets into the cord. The motor chain runs from the cord through the rootlets, anterior root, and spinal nerve out to a muscle. Add the posterior root ganglion and intervertebral foramen where they belong.

Then: Circle the box that holds the sensory cell bodies, and star the one nerve both chains travel in and write why it is called mixed.

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COMP

MODULE 5 · THE SPINAL CORD · LAB

Spinal cord internal anatomy

You should be able to: Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CROSS SECTION

Draw a cross section of a thoracic segment of the cord as it looks on the slide or model, with the gray butterfly inside the white matter. Label the posterior, lateral, and anterior horns, the gray commissure, the central canal, the anterior white commissure, and the anterior, lateral, and posterior white columns.

Then: Write beside each horn, in a few words, what it holds, and write tracts inside one white column.

☐ **B** LANDMARK TABLE

Draw a table with these down the side: the posterior, lateral, and anterior horns, the gray commissure, the anterior white commissure, the central canal, and the anterior, lateral, and posterior columns. Across the top: gray or white matter, how to find it on the section, and what it holds or connects. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a cervical and a thoracic section side by side and circle the lateral horn on the one that has it.

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COMP

MODULE 5 · THE PERIPHERAL NERVOUS SYSTEM · LECTURE AND LAB

Nerve connective tissue wrappings

You should be able to: Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps.

PICK ONE AND DO THAT ONE

☐ **A** LABELED NERVE CROSS SECTION

Draw a cross section of a peripheral nerve as it looks on the slide. Show the whole nerve with several fascicles inside it, then enlarge one fascicle so you can see a few axons. Label an axon, a fascicle, the endoneurium, the perineurium, and the epineurium.

Then: Color each wrapping a different shade and write beside each one the level it wraps: one axon, one fascicle, or the whole nerve.

☐ **B** WRAPPING TABLE

Draw a table with the three wrappings down the side: endoneurium, perineurium, and epineurium. Across the top: inner, middle, or outer layer, the level it wraps (axon, fascicle, or whole nerve), and a tiny sketch of that level with the wrapping drawn around it. Use a word or a tiny sketch in each cell.

Then: Below the table, write how the perineurium and epineurium help a surgeon repair a cut nerve.

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COMP

MODULE 5 · THE PERIPHERAL NERVOUS SYSTEM · LECTURE AND LAB

Spinal nerves and rami

You should be able to: State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SPINAL NERVE

Draw a cross section of the spinal cord with one spinal nerve leaving through the intervertebral foramen. Show it splitting into the posterior ramus, the anterior ramus, and the meningeal branch, with the rami communicantes linking the anterior ramus to a sympathetic trunk ganglion. Label each branch and draw an arrow toward the territory it supplies.

Then: Beside the cord, write the number of spinal nerve pairs in each region, cervical through coccygeal, and the total.

☐ **B** COUNT AND TERRITORY TABLE

Draw a table in two parts. In the top part, list the five cord regions down the side with the number of spinal nerve pairs in each, and the total. In the bottom part, list the posterior ramus, anterior ramus, meningeal branch, and rami communicantes, with the territory each supplies. Use a word or a tiny sketch in each cell.

Then: Star the ramus that forms the plexuses, and write the name and spinal levels of the thoracic nerves that do not form one.

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COMP

MODULE 5 · THE PERIPHERAL NERVOUS SYSTEM · LECTURE AND LAB

Sensory receptor classification

You should be able to: Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SKIN BLOCK

Draw a block of skin (epidermis, dermis, hypodermis) with one hair, and a skeletal muscle with its tendon underneath. Place and label free nerve endings, a tactile disc, a hair follicle receptor, a tactile corpuscle, a lamellar corpuscle, a bulbous corpuscle, a muscle spindle, and a tendon organ where the notes put them. Write what each one detects.

Then: Beside the drawing, write the three ways receptors are classified, and tag each receptor you drew as an exteroceptor or a proprioceptor.

☐ **B** RECEPTOR TABLE

Draw a table with the nine named receptors down the side, from free nerve endings to joint kinesthetic receptors. Across the top: where it sits, what it detects, and its class by stimulus (such as mechanoreceptor or nociceptor). Use a word or a tiny sketch in each cell.

Then: Below the table, list the three classes by location and the two by structure, and write one receptor from your table beside each class where one fits.

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MODULE 5 · THE CRANIAL NERVES · LECTURE AND LAB

Cranial nerves

You should be able to: Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen.

PICK ONE AND DO THAT ONE

☐ **A** LABELED INFERIOR VIEW

Draw the inferior view of the brain and brainstem. Draw and label all twelve cranial nerves, I to XII, where each one attaches, along with the olfactory bulb and tract, optic chiasm, optic tract, and pontomedullary junction. Beside each nerve, write S, M, or B for sensory, motor, or both.

Then: Write the skull foramen at the end of each nerve, and circle the three openings that carry more than one nerve.

☐ **B** CRANIAL NERVE TABLE

Draw a table with CN I to CN XII down the side. Across the top: name, class (sensory, motor, or mixed), primary function, and skull foramen. Write the names in order from memory first, using the memory sentence from the notes, then fill the other columns. Use a word or a tiny sketch in each cell.

Then: Color the rows that pass through the superior orbital fissure, the internal acoustic meatus, and the jugular foramen, using one color per opening.

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MODULE 5 · THE NERVE PLEXUSES · LECTURE AND LAB

Nerve plexuses

You should be able to: Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged.

PICK ONE AND DO THAT ONE

☐ **A** LABELED PLEXUS MAP

Draw the spinal cord with the spinal nerves leaving one side, grouped and numbered by region. Bracket and label the cervical, brachial, lumbar, sacral, and coccygeal plexuses, and write the spinal nerves that form each one. Draw the intercostal nerves running straight out from the thoracic region without forming a plexus.

Then: On a small body outline beside it, shade in matching colors the region that loses movement and sensation when each plexus is damaged.

☐ **B** DAMAGE TABLE

Draw a table with the five plexuses down the side. Across the top: the spinal nerves that form it, the region of the body it serves, and what a patient loses if it is damaged. Use a word or a tiny sketch in each cell.

Then: Below the table, draw the phrenic nerve leaving its plexus to reach the diaphragm, and write what happens to breathing if it is damaged.

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COMP

MODULE 5 · THE AUTONOMIC NERVOUS SYSTEM · LECTURE

Autonomic two-neuron pathway

You should be able to: Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector.

PICK ONE AND DO THAT ONE

☐ **A** SIDE BY SIDE PATHWAYS

Draw a somatic pathway and an autonomic pathway side by side, each starting in the spinal cord. Show the somatic one as a single neuron running to skeletal muscle. Show the autonomic one as a preganglionic neuron, an autonomic ganglion, and a postganglionic neuron reaching an effector. Label the cell bodies, the myelinated and unmyelinated axons, and the synapse.

Then: Write the three autonomic effectors at the end of the autonomic pathway, and write voluntary or involuntary under each pathway.

☐ **B** COMPARE TABLE

Draw a table with five features down the side: effector, control, number of motor neurons, effect on the effector, and awareness. Across the top: somatic nervous system and autonomic nervous system. Use a word or a tiny sketch in each cell.

Then: Below the table, draw the four-step autonomic route as a chain of boxes with arrows, from the preganglionic neuron to the effector tissue.

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COMP

MODULE 5 · THE AUTONOMIC NERVOUS SYSTEM · LECTURE

Autonomic ganglia and divisions

You should be able to: Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses.

PICK ONE AND DO THAT ONE

☐ **A** LABELED GANGLIA MAP

Draw the brainstem and spinal cord down the middle with the sympathetic trunk beside the vertebral column. Bracket where each division leaves the CNS. Draw and label a sympathetic trunk ganglion, the prevertebral ganglia near the large abdominal arteries, and a terminal ganglion in an organ wall. Add the cardiac plexus on the heart, the celiac plexus on the abdominal organs, and the hypogastric plexus on the pelvic organs.

Then: Draw a small piece of gut wall and place the myenteric and submucosal plexuses between the layers where each one sits.

☐ **B** COMPARE TABLE

Draw a table with sympathetic and parasympathetic across the top. Down the side: other name, origin in the CNS, location of its ganglia, the named ganglia that belong to it, and overall role. Use a word or a tiny sketch in each cell.

Then: Below the table, list the autonomic plexuses with what each supplies, and the two enteric plexuses with the gut layers each sits between.

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